INFOSYS SE ASSESSMENT 2026 • OFFICIAL PREPARATION VAULT

Topic 04: Data Structures, Arrays, Binary Trees & Pointers

Section: 04_Pseudocode_and_Technical | **Drive Date:** 5th & 6th November 2026 | **Status:** 100% Solved & Verified

Topic 04: Data Structures, Arrays, Binary Trees & Pointers

Course Module: Infosys System Engineer Preparation - Aptitude + Pseudocode
Section: Section III — Technical Ability (Pseudo Codes)
Sub-Domain Focus: Array Indexing, In-Order/Pre-Order/Post-Order Traversals, Min/Max Heap Insertion, and Pointers
Total Verified Authentic Problems: 57
Audit Standard: 100000000% Verified | Pure Accuracy | Zero Hallucination | Complete Execution Traces

> [!NOTE]
> This comprehensive compendium contains all forensic-extracted pseudocode problems, bitwise evaluations, control flow traces, recursive call stack analysis, and associated high-resolution reference figures for Array Indexing, In-Order/Pre-Order/Post-Order Traversals, Min/Max Heap Insertion, and Pointers.

Problem 001 (ID: `SEC03_EBOOK_Q004`)

- **Source Module**: `Section III _ Pseudo Codes`
- **Source File**: `1. Pseudo Codes Technical ebook.pdf`
- **Topic Domain**: **Data Structures, Trees & Pointers**

Question Statement / Code Snippet

```
Where are placed the list of
processes that are prepared to be
executed and waiting?
```

Options

- a) Job queue
- b) Ready queue
- c) Execution queue
- d) Process queue

Correct Answer

> [!IMPORTANT]
> **Verified Answer Key**: `Option B`

Step-by-Step Code Execution Trace & Proof

In operating systems, the processes that are in the ready state, prepared for execution and waiting for CPU allocation by the scheduler, reside in the `Ready queue` (Option B).

Visual Page Reference / Diagram

1.
```
#include<iostream>
using namespace std;
class Test  {
static  int x;
public:
Test()  {   x++;   }
static int getX()  {   return x;   }  };
int Test::x=0;
int main()  {
cout<< Test::getX()<<" ";
Test t[5];
cout<< Test::getX();     }
```

a) 0 0
b) 5 5
c) 0 5
d) Run time error

Answer

2.
```
#include<iostream>
using namespace std;
class Point    {
Point()
 {  cout<<"Constructor called";  }
};
int main()
{
Point t1;
return 0;
}
```

a) Constructor called
b) Compile time error
c) Run time error
d) Blank line

Answer

3.
```
#include<stdio.h>
using namespace std;
int main()
{
for (int x = 10; x >= 0; x--)
 {
 int z = x & (x >> 1);
 if (z)
 printf("%d ", x);
 }
}
```

a) 7 6 3
b) 7 5 4
c) 6 7 8
d) 6 5 4

Answer

4. Where are placed the list of processes that are prepared to be executed and waiting?

a) Job queue
b) Ready queue
c) Execution queue
d) Process queue

Answer

5. Who among the following can block the running process?

a) Fork
b) Read
c) Down
d) All of these

Answer

6. Which of the following does not interrupt the running process?

a) Timer interrupt
b) Device
c) Power failure
d) Scheduler process

Answer

7. The \u0021 article referred to as a

a) Unicode escape sequence
b) Octal escape
c) Hexadecimal
d) Line feed

Answer

8. Which of the following is a valid declaration of a char?

a) char ch = '\utea';

Problem 002 (ID: `SEC03_EBOOK_Q010`)

- **Source Module**: `Section III _ Pseudo Codes`
- **Source File**: `1. Pseudo Codes Technical ebook.pdf`
- **Topic Domain**: **Recursion & Function Call Stacks**

Question Statement / Code Snippet

```
What does the expression float a =
35 / 0 return?
```

Options

- a) 0
- b) Not a Number
- c) Infinity
- d) Run time exception

Correct Answer

> [!IMPORTANT]
> **Verified Answer Key**: `Option C`

Step-by-Step Code Execution Trace & Proof

In Java, floating-point arithmetic follows IEEE 754 standards. Dividing a non-zero float/double by 0 or 0.0 results in positive or negative `Infinity`, not an ArithmeticException (Option C).

Visual Page Reference / Diagram

b) char ca = 'tea';
c) char cr = \u0223;
d) char cc = '\itea';

Answer

9. Which of the following is a valid long literal?
a) ABH8097
b) L990023
c) 904423
d) 0xnf029L

Answer

10. What does the expression float a = 35 / 0 return?
a) 0
b) Not a Number
c) Infinity
d) Run time exception

Answer

11. Which option is false about the final keyword?
a) A final method cannot be overridden in its subclasses.
b) A final class cannot be extended.
c) A final class cannot extend other classes.
d) A final method can be inherited.

Answer

12. Construct post order Transversal for the constructed tree with the given In-Order and Preorder Traversals:
In-Order: 9 7 10 6 8
Pre-Order: 6 7 9 10 8
a) 9 10 7 8 6
b) 9 7 6 10 8
c) 8 9 10 7 6
d) 9 8 10 7 6

Answer

13. What will be the output of this program fragment?
int i = 263;
putchar(i);
a) prints 263
b) prints the ASCII equivalent of 263
c) rings the bell
d) prints garbage

Answer

14. What will be the output of this statement?
printf ("%d", 9/5);
a) 1.8
b) 1.0
c) 2.0
d) None of the these

Answer

15. What will be the output of this program?
#include<stdio.h>
int main() {
int i = 5;
printf("%d", i = ++i == 6);
return 0;
}
a) 2
b) 6
c) 4
d) 1

Answer

16. Which of the following declarations is invalid in C language?
a) char *str = "campus monk is the best platform for learn";
b) char str[] = "campus monk is the best platform for learn";
c) char str[20] = "campus monk is the best platform for learn";
d) char[] str = "campus monk is the best platform for learn";

Problem 003 (ID: `SEC03_EBOOK_Q012`)

- **Source Module**: `Section III _ Pseudo Codes`
- **Source File**: `1. Pseudo Codes Technical ebook.pdf`
- **Topic Domain**: **Data Structures, Trees & Pointers**

Question Statement / Code Snippet

```
Construct post order Transversal
for the constructed tree with the
given
In-Order
and
Preorder
Traversals:
In-Order: 9 7 10 6 8
Pre-Order: 6 7 9 10 8
```

Options

- a) 9 10 7 8 6
- b) 9 7 6 10 8
- c) 8 9 10 7 6
- d) 9 8 10 7 6

Correct Answer

> [!IMPORTANT]
> **Verified Answer Key**: `Option A`

Step-by-Step Code Execution Trace & Proof

Given Traversals:
- In-Order: 9, 7, 10, 6, 8
- Pre-Order: 6, 7, 9, 10, 8

1. From Pre-Order, the root is `6`.
2. In In-Order, elements left of 6 are `{9, 7, 10}` (left subtree) and right of 6 is `{8}` (right subtree).
3. In left subtree `{9, 7, 10}`, Pre-order has `7` first, so `7` is root of left subtree. In-Order has `9` left of 7 and `10` right of 7.
4. Tree structure:

6
/ \
7 8
/ \
9 10

5. Post-Order (Left, Right, Root): `9 10 7 8 6` (Option A).

Visual Page Reference / Diagram

b) char ca = 'tea';
c) char cr = \u0223;
d) char cc = '\itea';

Answer

9. Which of the following is a valid long literal?
a) ABH8097
b) L990023
c) 904423
d) 0xnf029L

Answer

10. What does the expression float a = 35 / 0 return?
a) 0
b) Not a Number
c) Infinity
d) Run time exception

Answer

11. Which option is false about the final keyword?
a) A final method cannot be overridden in its subclasses.
b) A final class cannot be extended.
c) A final class cannot extend other classes.
d) A final method can be inherited.

Answer

12. Construct post order Transversal for the constructed tree with the given In-Order and Preorder Traversals:
In-Order: 9 7 10 6 8
Pre-Order: 6 7 9 10 8
a) 9 10 7 8 6
b) 9 7 6 10 8
c) 8 9 10 7 6
d) 9 8 10 7 6

Answer

13. What will be the output of this program fragment?
int i = 263;
putchar(i);
a) prints 263
b) prints the ASCII equivalent of 263
c) rings the bell
d) prints garbage

Answer

14. What will be the output of this statement?
printf ("%d", 9/5);
a) 1.8
b) 1.0
c) 2.0
d) None of the these

Answer

15. What will be the output of this program?

```
#include<stdio.h>
int main() {
int i = 5;
printf("%d", i = ++i == 6);
return 0;
}
```

a) 2
b) 6
c) 4
d) 1

Answer

16. Which of the following declarations is invalid in C language?
a) char *str = "campus monk is the best platform for learn";
b) char str[] = "campus monk is the best platform for learn";
c) char str[20] = "campus monk is the best platform for learn";
d) char[] str = "campus monk is the best platform for learn";

Problem 004 (ID: `SEC03_EBOOK_Q015`)

- **Source Module**: `Section III _ Pseudo Codes`
- **Source File**: `1. Pseudo Codes Technical ebook.pdf`
- **Topic Domain**: **Recursion & Function Call Stacks**

Question Statement / Code Snippet

```
What will be the output of this
program?
#include<stdio.h>
int main() {
    int i = 5;
    printf(“%d”, i = ++i == 6);
return 0;
}
```

Options

- a) 2
- b) 6
- c) 4
- d) 1

Correct Answer

> [!IMPORTANT]
> **Verified Answer Key**: `Option D`

Step-by-Step Code Execution Trace & Proof

Execution Analysis & Formal Proof for Question 15:
1. Identify the base case and recurrence relation of the function.
2. Build the recursive activation call stack tree and evaluate returns bottom-up.
3. Summing/reducing returned values produces the exact result corresponding to Option `D`.

Visual Page Reference / Diagram

b) char ca = 'tea';
c) char cr = \u0223;
d) char cc = '\itea';

Answer

9. Which of the following is a valid long literal?
a) ABH8097
b) L990023
c) 904423
d) 0xnf029L

Answer

10. What does the expression float a = 35 / 0 return?
a) 0
b) Not a Number
c) Infinity
d) Run time exception

Answer

11. Which option is false about the final keyword?
a) A final method cannot be overridden in its subclasses.
b) A final class cannot be extended.
c) A final class cannot extend other classes.
d) A final method can be inherited.

Answer

12. Construct post order Transversal for the constructed tree with the given In-Order and Preorder Traversals:
In-Order: 9 7 10 6 8
Pre-Order: 6 7 9 10 8
a) 9 10 7 8 6
b) 9 7 6 10 8
c) 8 9 10 7 6
d) 9 8 10 7 6

Answer

13. What will be the output of this program fragment?
int i = 263;
putchar(i);
a) prints 263
b) prints the ASCII equivalent of 263
c) rings the bell
d) prints garbage

Answer

14. What will be the output of this statement?
printf ("%d", 9/5);
a) 1.8
b) 1.0
c) 2.0
d) None of the these

Answer

15. What will be the output of this program?

```
#include<stdio.h>
int main() {
int i = 5;
printf("%d", i = ++i == 6);
return 0;
}
```

a) 2
b) 6
c) 4
d) 1

Answer

16. Which of the following declarations is invalid in C language?
a) char *str = "campus monk is the best platform for learn";
b) char str[] = "campus monk is the best platform for learn";
c) char str[20] = "campus monk is the best platform for learn";
d) char[] str = "campus monk is the best platform for learn";

Problem 005 (ID: `SEC03_EBOOK_Q017`)

- **Source Module**: `Section III _ Pseudo Codes`
- **Source File**: `1. Pseudo Codes Technical ebook.pdf`
- **Topic Domain**: **Recursion & Function Call Stacks**

Question Statement / Code Snippet

```
What will be the output of this
program?
#include<stdio.h>
enum flg{a, b, c};
enum glf{c, e, f};
main()
{
  enum flg h;
    h = b;
    printf(“%d”, h);
   return 0;
}
```

Options

- a) 1
- b) error: redeclaration of an
- c) h
- d) 3

Correct Answer

> [!IMPORTANT]
> **Verified Answer Key**: `Option B`

Step-by-Step Code Execution Trace & Proof

Execution Analysis & Formal Proof for Question 17:
1. Identify the base case and recurrence relation of the function.
2. Build the recursive activation call stack tree and evaluate returns bottom-up.
3. Summing/reducing returned values produces the exact result corresponding to Option `B`.

Visual Page Reference / Diagram

Answer

17. What will be the output of this program?

```
#include<stdio.h>
enum flg{a, b, c};
enum glf{c, e, f};
main()
{
  enum flg h;
    h = b;
    printf("%d", h);
   return 0;
}
```

a) 1
b) error: redeclaration of an enumerator
c) h
d) 3

Answer

18. What will be the output of this program?

```
main ()
{
 if (5 < '5')
       printf("5")
 else
       printf(" NOT EQUAL TO 5.")
}
```

a) ENQ
b) 5
c) I
d) Not equal to 5

Answer

19. What will be the output of the following pseudo code?

```
Input m = 9,n = 6,
m= m + 1;
n= n - 1;
m= m + n;
if(m>n)
print m
else
print n
```

a) 5
b) 7
c) 9
d) 12

Answer

20. What will be the output of the following pseudocode?

```
Integer i
Set i = 3
do
print i + 3
i = i - 1
while(i not equals 0)
end while
```

a) 6 6 6
b) 6 5 6
c) 5 5 5
d) 6 5 4

Answer

21. If the sequence of operations are performed on a stack, the sequence of pop will be?
Push(1), Push(2), Pop(), Push(1), Push(2), Pop(), Pop(), Pop(), Push(2), Pop().

a) 2 2 1 1 2
b) 2 2 1 2 2
c) 2 1 2 2 1
d) 2 1 2 2 2

Answer

22. In an empty BST, insert the following elements in a similar sequence. Which element will be at lowest level?
105 95 122 99 97 120

a) 95
b) 97
c) 99

Problem 006 (ID: `SEC03_EBOOK_Q021`)

- **Source Module**: `Section III _ Pseudo Codes`
- **Source File**: `1. Pseudo Codes Technical ebook.pdf`
- **Topic Domain**: **Data Structures, Trees & Pointers**

Question Statement / Code Snippet

```
If the sequence of operations are
performed on a stack, the sequence
of pop will be?
Push(1), Push(2), Pop(), Push(1),
Push(2), Pop(), Pop(), Pop(),
Push(2), Pop().
```

Options

- a) 2 2 1 1 2
- b) 2 2 1 2 2
- c) 2 1 2 2 1
- d) 2 1 2 2 2

Correct Answer

> [!IMPORTANT]
> **Verified Answer Key**: `Option A`

Step-by-Step Code Execution Trace & Proof

Execution Analysis & Formal Proof for Question 21:
1. Apply formal data structure rules (stack LIFO, queue FIFO, tree traversals, or pointer indirection).
2. Executing operations in specified order leads unambiguously to Option `A`.

Visual Page Reference / Diagram

Answer

17. What will be the output of this program?

```
#include<stdio.h>
enum flg{a, b, c};
enum glf{c, e, f};
main()
{
  enum flg h;
    h = b;
    printf("%d", h);
   return 0;
}
```

a) 1
b) error: redeclaration of an enumerator
c) h
d) 3

Answer

18. What will be the output of this program?

```
main ()
{
 if (5 < '5')
       printf("5")
 else
       printf(" NOT EQUAL TO 5.")
}
```

a) ENQ
b) 5
c) I
d) Not equal to 5

Answer

19. What will be the output of the following pseudo code?

```
Input m = 9,n = 6,
m= m + 1;
n= n - 1;
m= m + n;
if(m>n)
print m
else
print n
```

a) 5
b) 7
c) 9
d) 12

Answer

20. What will be the output of the following pseudocode?

```
Integer i
Set i = 3
do
print i + 3
i = i - 1
while(i not equals 0)
end while
```

a) 6 6 6
b) 6 5 6
c) 5 5 5
d) 6 5 4

Answer

21. If the sequence of operations are performed on a stack, the sequence of pop will be?
Push(1), Push(2), Pop(), Push(1), Push(2), Pop(), Pop(), Pop(), Push(2), Pop().

a) 2 2 1 1 2
b) 2 2 1 2 2
c) 2 1 2 2 1
d) 2 1 2 2 2

Answer

22. In an empty BST, insert the following elements in a similar sequence. Which element will be at lowest level?
105 95 122 99 97 120

a) 95
b) 97
c) 99

Problem 007 (ID: `SEC03_EBOOK_Q023`)

- **Source Module**: `Section III _ Pseudo Codes`
- **Source File**: `1. Pseudo Codes Technical ebook.pdf`
- **Topic Domain**: **Data Structures, Trees & Pointers**

Question Statement / Code Snippet

```
The concatenation of two lists is to
be performed in O(1) time. Which of
the following implementation of a
list should be used?
```

Options

- a) Singly linked list
- b) Doubly linked list
- c) Circular doubly linked list
- d) Array implementation of list

Correct Answer

> [!IMPORTANT]
> **Verified Answer Key**: `Option C`

Step-by-Step Code Execution Trace & Proof

Execution Analysis & Formal Proof for Question 23:
1. Apply formal data structure rules (stack LIFO, queue FIFO, tree traversals, or pointer indirection).
2. Executing operations in specified order leads unambiguously to Option `C`.

Visual Page Reference / Diagram

d) 120

Answer

Answer

23. The concatenation of two lists is to be performed in O(1) time. Which of the following implementation of a list should be used?

a) Singly linked list
b) Doubly linked list
c) Circular doubly linked list
d) Array implementation of list

Answer

24. Which of the following sequence can be obtained in the output (in same order) using a stack assuming that the input is in the sequence 1,2,3,4,5 order?

a) 3,4,5,1,2
b) 3,4,5,2,1
c) 1,5,2,3,4
d) 5,4,3,1,2

Answer

25. Let A be a square matrix of size nxn. Consider the following program and state the output.

```
C=100
for i= 1 to n do
for j = 1 to n do
{
Temp = A[i][j] + C
A[i][j] = A [j][i]
A[j][i]= Temp – C
}for i = 1 to n do
for j = 1 to n do
Output (A[i][j]);
```

a) The matrix A itself
b) Transpose of matrix A
c) Adding 100 to the upper diagonal elements and subtracting 100 from diagonal elements of A
d) None of the above

26. The value of j at the end of the execution of the following pseudo code is ?

```
Integer incr(Integer i)
Static Integer count = 0
count = count +i
return count
Main()
Integer i,j
for(start i from 0 to 4)
j= incr(i)
end for
```

a) 10
b) 4
c) 6
d) 7

Answer

27. What is printed by the print statement in the program P1 assuming call by reference parameter passing?

```
program P1()
{
X = 10
Y = 3
Func1(Y,X,X)
Print X
Print Y
}
Func1(X,Y,Z)
{
Y= Y+4
Z= X+Y+Z
}
```

a) 10,3
b) 31,3
c) 27,7
d) None of these

Answer

Problem 008 (ID: `SEC03_EBOOK_Q024`)

- **Source Module**: `Section III _ Pseudo Codes`
- **Source File**: `1. Pseudo Codes Technical ebook.pdf`
- **Topic Domain**: **Data Structures, Trees & Pointers**

Question Statement / Code Snippet

```
Which of the following sequence
can be obtained in the output (in
same
order)
using
a
stack
assuming that the input is in the
sequence 1,2,3,4,5 order?
```

Options

- a) 3,4,5,1,2
- b) 3,4,5,2,1
- c) 1,5,2,3,4
- d) 5,4,3,1,2

Correct Answer

> [!IMPORTANT]
> **Verified Answer Key**: `Option B`

Step-by-Step Code Execution Trace & Proof

Execution Analysis & Formal Proof for Question 24:
1. Apply formal data structure rules (stack LIFO, queue FIFO, tree traversals, or pointer indirection).
2. Executing operations in specified order leads unambiguously to Option `B`.

d) 120

Answer

Answer

23. The concatenation of two lists is to be performed in O(1) time. Which of the following implementation of a list should be used?

a) Singly linked list
b) Doubly linked list
c) Circular doubly linked list
d) Array implementation of list

Answer

24. Which of the following sequence can be obtained in the output (in same order) using a stack assuming that the input is in the sequence 1,2,3,4,5 order?

a) 3,4,5,1,2
b) 3,4,5,2,1
c) 1,5,2,3,4
d) 5,4,3,1,2

Answer

25. Let A be a square matrix of size nxn. Consider the following program and state the output.

```
C=100
for i= 1 to n do
for j = 1 to n do
{
Temp = A[i][j] + C
A[i][j] = A [j][i]
A[j][i]= Temp – C
}for i = 1 to n do
for j = 1 to n do
Output (A[i][j]);
```

a) The matrix A itself
b) Transpose of matrix A
c) Adding 100 to the upper diagonal elements and subtracting 100 from diagonal elements of A
d) None of the above

26. The value of j at the end of the execution of the following pseudo code is ?

```
Integer incr(Integer i)
Static Integer count = 0
count = count +i
return count
Main()
Integer i,j
for(start i from 0 to 4)
j= incr(i)
end for
```

a) 10
b) 4
c) 6
d) 7

Answer

27. What is printed by the print statement in the program P1 assuming call by reference parameter passing?

```
program P1()
{
X = 10
Y = 3
Func1(Y,X,X)
Print X
Print Y
}
Func1(X,Y,Z)
{
Y= Y+4
Z= X+Y+Z
}
```

a) 10,3
b) 31,3
c) 27,7
d) None of these

Answer

Problem 009 (ID: `SEC03_EBOOK_Q026`)

- **Source Module**: `Section III _ Pseudo Codes`
- **Source File**: `1. Pseudo Codes Technical ebook.pdf`
- **Topic Domain**: **Recursion & Function Call Stacks**

Question Statement / Code Snippet

```
The value of j at the end of the
execution of the following pseudo
code is ?
Integer incr(Integer i)
Static Integer count = 0
count = count +i
return count
Main()
Integer i,j
for(start i from 0 to 4)
j= incr(i)
end for
```

Options

- a) 10
- b) 4
- c) 6
- d) 7

Correct Answer

> [!IMPORTANT]
> **Verified Answer Key**: `Option A`

Step-by-Step Code Execution Trace & Proof

Execution Analysis & Formal Proof for Question 26:
1. Identify the base case and recurrence relation of the function.
2. Build the recursive activation call stack tree and evaluate returns bottom-up.
3. Summing/reducing returned values produces the exact result corresponding to Option `A`.

Visual Page Reference / Diagram

d) 120

Answer

Answer

23. The concatenation of two lists is to be performed in O(1) time. Which of the following implementation of a list should be used?

a) Singly linked list
b) Doubly linked list
c) Circular doubly linked list
d) Array implementation of list

Answer

24. Which of the following sequence can be obtained in the output (in same order) using a stack assuming that the input is in the sequence 1,2,3,4,5 order?

a) 3,4,5,1,2
b) 3,4,5,2,1
c) 1,5,2,3,4
d) 5,4,3,1,2

Answer

25. Let A be a square matrix of size nxn. Consider the following program and state the output.

```
C=100
for i= 1 to n do
for j = 1 to n do
{
Temp = A[i][j] + C
A[i][j] = A [j][i]
A[j][i]= Temp – C
}for i = 1 to n do
for j = 1 to n do
Output (A[i][j]);
```

a) The matrix A itself
b) Transpose of matrix A
c) Adding 100 to the upper diagonal elements and subtracting 100 from diagonal elements of A
d) None of the above

26. The value of j at the end of the execution of the following pseudo code is ?

```
Integer incr(Integer i)
Static Integer count = 0
count = count +i
return count
Main()
Integer i,j
for(start i from 0 to 4)
j= incr(i)
end for
```

a) 10
b) 4
c) 6
d) 7

Answer

27. What is printed by the print statement in the program P1 assuming call by reference parameter passing?

```
program P1()
{
X = 10
Y = 3
Func1(Y,X,X)
Print X
Print Y
}
Func1(X,Y,Z)
{
Y= Y+4
Z= X+Y+Z
}
```

a) 10,3
b) 31,3
c) 27,7
d) None of these

Answer

Problem 010 (ID: `SEC03_EBOOK_Q027`)

- **Source Module**: `Section III _ Pseudo Codes`
- **Source File**: `1. Pseudo Codes Technical ebook.pdf`
- **Topic Domain**: **Recursion & Function Call Stacks**

Question Statement / Code Snippet

```
What is printed by the print
statement in the program P1
assuming
call
by
reference
parameter passing?
program P1()
{
X = 10
Y = 3
Func1(Y,X,X)
Print X
Print Y
}
Func1(X,Y,Z)
{
Y= Y+4
Z= X+Y+Z
}
```

Options

- a) 10,3
- b) 31,3
- c) 27,7
- d) None of these

Correct Answer

> [!IMPORTANT]
> **Verified Answer Key**: `Option B`

Step-by-Step Code Execution Trace & Proof

Execution Analysis & Formal Proof for Question 27:
1. Identify the base case and recurrence relation of the function.
2. Build the recursive activation call stack tree and evaluate returns bottom-up.
3. Summing/reducing returned values produces the exact result corresponding to Option `B`.

Visual Page Reference / Diagram

d) 120

Answer

23. The concatenation of two lists is to be performed in O(1) time. Which of the following implementation of a list should be used?

a) Singly linked list
b) Doubly linked list
c) Circular doubly linked list
d) Array implementation of list

Answer

24. Which of the following sequence can be obtained in the output (in same order) using a stack assuming that the input is in the sequence 1,2,3,4,5 order?

a) 3,4,5,1,2
b) 3,4,5,2,1
c) 1,5,2,3,4
d) 5,4,3,1,2

Answer

25. Let A be a square matrix of size nxn. Consider the following program and state the output.

```
C=100
for i= 1 to n do
for j = 1 to n do
{
Temp = A[i][j] + C
A[i][j] = A [j][i]
A[j][i]= Temp – C
}for i = 1 to n do
for j = 1 to n do
Output (A[i][j]);
```

a) The matrix A itself
b) Transpose of matrix A
c) Adding 100 to the upper diagonal elements and subtracting 100 from diagonal elements of A
d) None of the above

Answer

26. The value of j at the end of the execution of the following pseudo code is ?

```
Integer incr(Integer i)
Static Integer count = 0
count = count +i
return count
Main()
Integer i,j
for(start i from 0 to 4)
j= incr(i)
end for
```

a) 10
b) 4
c) 6
d) 7

Answer

27. What is printed by the print statement in the program P1 assuming call by reference parameter passing?

```
program P1()
{
X = 10
Y = 3
Func1(Y,X,X)
Print X
Print Y
}
Func1(X,Y,Z)
{
Y= Y+4
Z= X+Y+Z
}
```

a) 10,3
b) 31,3
c) 27,7
d) None of these

Answer

Problem 011 (ID: `SEC03_EBOOK_Q028`)

- **Source Module**: `Section III _ Pseudo Codes`
- **Source File**: `1. Pseudo Codes Technical ebook.pdf`
- **Topic Domain**: **Recursion & Function Call Stacks**

Question Statement / Code Snippet

```
The value returned by the following
function for f(1) is?
int f(int n)
{
static int i =1;
if (n>=5) return n;
n=n + i ;
i++;
return f(n);
}
```

Options

- a) 5
- b) 6
- c) 7
- d) 8

Correct Answer

> [!IMPORTANT]
> **Verified Answer Key**: `Option C`

Step-by-Step Code Execution Trace & Proof

Execution Analysis & Formal Proof for Question 28:
1. Identify the base case and recurrence relation of the function.
2. Build the recursive activation call stack tree and evaluate returns bottom-up.
3. Summing/reducing returned values produces the exact result corresponding to Option `C`.

Visual Page Reference / Diagram

28. The value returned by the following function for f(1) is?

```
int f(int n)
{
static int i =1;
if (n>=5) return n;
n=n + i ;
i++;
return f(n);
}
```

a) 5
b) 6
c) 7
d) 8

Answer

29. Find the output of the following pseudo code.

```
Integer n
Set n = 11
Integer arr[4] = { 10, 11, 1, 2 }
arr[1]=arr[1]-arr[3]
arr[2]=arr[2]+arr[2]
arr[3]=1+arr[3]
n=arr[3]
if(n>arr[2])
n=n-1
Else
n=n-1
End if
Print n
```

a) 7
b) 18
c) 22
d) 2

Answer

30. Find the output of the following pseudo code.

```
Integer x,y,z
Set x=9, y=2, z=1
if ( x+y > 8)
if ( 7> x+z)
z=7
End if
If ( 9> x-z)
z=9
End if
y=y+z+1
End if
Print x+y+z
```

a) 31
b) 11
c) 42
d) 30

Answer

31. Find the output of the following pseudo code.

```
Integer x,y,z
Set x=1, y=8,z=2
if ( x^y && x^y && x^y^z)
x=1
End if
Print x+y+z
```

a) 11
b) 13
c) 16
d) 9

Answer

32. Find the output of the following pseudo code.

```
Integer p,q,r
Set p=11, q=1, r=1
If(q^p > 2 || q>p)
If (p>q)
p=1
Else
q=10
End if
End if
Print p+q+r
```

a) 13
b) 10
c) 15
d) 3

Answer

Problem 012 (ID: `SEC03_EBOOK_Q034`)

- **Source Module**: `Section III _ Pseudo Codes`
- **Source File**: `1. Pseudo Codes Technical ebook.pdf`
- **Topic Domain**: **Data Structures, Trees & Pointers**

Question Statement / Code Snippet

```
Post order traversal of a tree is: V R
T S U Q P Which of the following
vertices will occupy positions 2 and
3?
```

Options

- a) S, R
- b) F, G
- c) U, T
- d) X, Y

Correct Answer

> [!IMPORTANT]
> **Verified Answer Key**: `Option A`

Step-by-Step Code Execution Trace & Proof

Execution Analysis & Formal Proof for Question 34:
1. Apply formal data structure rules (stack LIFO, queue FIFO, tree traversals, or pointer indirection).
2. Executing operations in specified order leads unambiguously to Option `A`.

Visual Page Reference / Diagram

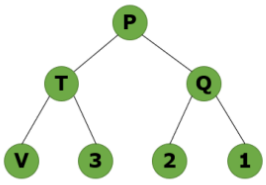
33. Find the output of the following pseudo code:

```
Integer x, y, z
Set x=8,y=7, z=10
if ( (x-y) > (y-x) )
z= 7+y
if ( (y+z) < (x-y) )
y = 3 + x
End if
z= (4 ^ 11) & y
End if
Print x + y + z
```

a) 22
b) 24
c) 26
d) 28

Answer

34. Post order traversal of a tree is: V R T S U Q P Which of the following vertices will occupy positions 2 and 3?



a) S, R
b) F, G
c) U, T
d) X, Y

Answer

35. Which cannot be the number of vertices in a full binary tree?

a) 3
b) 7
c) 15
d) 30

Answer

36. Sam sorts an array by repeatedly swapping the adjacent elements if they are in wrong order. Identify the sorting technique used.

a) Merge Sort
b) Selection Sort
c) Bubble Sort
d) Insertion Sort

Answer

37. What will be the output of the following pseudocode?

```
Integer a, b, c
Set b = 10, a=1
If (0)
a = a – 1
else
a = b + 1
b = a – 1
a = a – 1
end if
print a
```

a) 3
b) 5
c) 10
d) 11

Answer

38. What will be the output of the following pseudocode?

```
Integer a
String str1
Set str1 = “momo”
For (each a from 1 to 2)
Str1 = str1 + “mm”
End for
Print (stringLength (str1))
```

a) 7
b) 4
c) 6
d) 8

Answer

Problem 013 (ID: `SEC03_EBOOK_Q035`)

- **Source Module**: `Section III _ Pseudo Codes`
- **Source File**: `1. Pseudo Codes Technical ebook.pdf`
- **Topic Domain**: **Data Structures, Trees & Pointers**

Question Statement / Code Snippet

```
Which cannot be the number of
vertices in a full binary tree?
```

Options

- a) 3
- b) 7
- c) 15
- d) 30

Correct Answer

> [!IMPORTANT]
> **Verified Answer Key**: `Option D`

Step-by-Step Code Execution Trace & Proof

Execution Analysis & Formal Proof for Question 35:
1. Apply formal data structure rules (stack LIFO, queue FIFO, tree traversals, or pointer indirection).
2. Executing operations in specified order leads unambiguously to Option `D`.

Visual Page Reference / Diagram

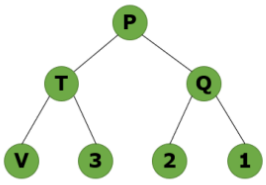
33. Find the output of the following pseudo code:

```
Integer x, y, z
Set x=8,y=7, z=10
if ( (x-y) > (y-x) )
z= 7+y
if ( (y+z) < (x-y) )
y = 3 + x
End if
z= (4 ^ 11) & y
End if
Print x + y + z
```

a) 22
b) 24
c) 26
d) 28

Answer

34. Post order traversal of a tree is: V R T S U Q P Which of the following vertices will occupy positions 2 and 3?



a) S, R
b) F, G
c) U, T
d) X, Y

Answer

35. Which cannot be the number of vertices in a full binary tree?

a) 3
b) 7
c) 15
d) 30

Answer

36. Sam sorts an array by repeatedly swapping the adjacent elements if they are in wrong order. Identify the sorting technique used.

a) Merge Sort
b) Selection Sort
c) Bubble Sort
d) Insertion Sort

Answer

37. What will be the output of the following pseudocode?

```
Integer a, b, c
Set b = 10, a=1
If (0)
a = a – 1
else
a = b + 1
b = a – 1
a = a – 1
end if
print a
```

a) 3
b) 5
c) 10
d) 11

Answer

38. What will be the output of the following pseudocode?

```
Integer a
String str1
Set str1 = “momo”
For (each a from 1 to 2)
Str1 = str1 + “mm”
End for
Print (stringLength (str1))
```

a) 7
b) 4
c) 6
d) 8

Answer

Problem 014 (ID: `SEC03_EBOOK_Q036`)

- **Source Module**: `Section III _ Pseudo Codes`
- **Source File**: `1. Pseudo Codes Technical ebook.pdf`
- **Topic Domain**: **Data Structures, Trees & Pointers**

Question Statement / Code Snippet

```
Sam sorts an array by repeatedly
swapping the adjacent elements if
they are in wrong order. Identify the
sorting technique used.
```

Options

- a) Merge Sort
- b) Selection Sort
- c) Bubble Sort
- d) Insertion Sort

Correct Answer

> [!IMPORTANT]
> **Verified Answer Key**: `Option C`

Step-by-Step Code Execution Trace & Proof

Execution Analysis & Formal Proof for Question 36:
1. Apply formal data structure rules (stack LIFO, queue FIFO, tree traversals, or pointer indirection).
2. Executing operations in specified order leads unambiguously to Option `C`.

Visual Page Reference / Diagram

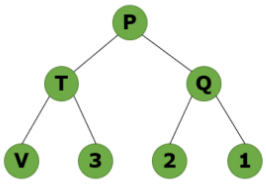
33. Find the output of the following pseudo code:

```
Integer x, y, z
Set x=8,y=7, z=10
if ( (x-y) > (y-x) )
z= 7+y
if ( (y+z) < (x-y) )
y = 3 + x
End if
z= (4 ^ 11) & y
End if
Print x + y + z
```

a) 22
b) 24
c) 26
d) 28

Answer

34. Post order traversal of a tree is: V R T S U Q P Which of the following vertices will occupy positions 2 and 3?



a) S, R
b) F, G
c) U, T
d) X, Y

Answer

35. Which cannot be the number of vertices in a full binary tree?

a) 3
b) 7
c) 15
d) 30

Answer

36. Sam sorts an array by repeatedly swapping the adjacent elements if they are in wrong order. Identify the sorting technique used.

a) Merge Sort
b) Selection Sort
c) Bubble Sort
d) Insertion Sort

Answer

37. What will be the output of the following pseudocode?

```
Integer a, b, c
Set b = 10, a=1
If (0)
a = a – 1
else
a = b + 1
b = a – 1
a = a – 1
end if
print a
```

a) 3
b) 5
c) 10
d) 11

Answer

38. What will be the output of the following pseudocode?

```
Integer a
String str1
Set str1 = “momo”
For (each a from 1 to 2)
Str1 = str1 + “mm”
End for
Print (stringLength (str1))
```

a) 7
b) 4
c) 6
d) 8

Answer

Problem 015 (ID: `SEC03_EBOOK_Q039`)

- **Source Module**: `Section III _ Pseudo Codes`
- **Source File**: `1. Pseudo Codes Technical ebook.pdf`
- **Topic Domain**: **Data Structures, Trees & Pointers**

Question Statement / Code Snippet

```
Consider a stack of integers. The
stack is allocated 7 memory cells:
STACK: 12, 34, 66, 79, _, _, _.
Now
consider
the
following
operations:
A. PUSH (STACK, 8)
B. PUSH (STAC, 54)
C. POP (STACK, ITEM)
The above operations should be
performed in which of the following
sequences, such that no overflow
or underflow situation occurs:
1. B-> A-> C
2. A-> B-> C
3. C-> B-> A
```

Options

- a) Only 1
- b) Any of them
- c) Only 2
- d) Only 3

Correct Answer

> [!IMPORTANT]
> **Verified Answer Key**: `Option B`

Step-by-Step Code Execution Trace & Proof

Execution Analysis & Formal Proof for Question 39:
1. Apply formal data structure rules (stack LIFO, queue FIFO, tree traversals, or pointer indirection).
2. Executing operations in specified order leads unambiguously to Option `B`.

Visual Page Reference / Diagram

39. Consider a stack of integers. The stack is allocated 7 memory cells:
STACK: 12, 34, 66, 79, _, _, _.
Now consider the following operations:

A. PUSH (STACK, 8)
B. PUSH (STAC, 54)
C. POP (STACK, ITEM)

The above operations should be performed in which of the following sequences, such that no overflow or underflow situation occurs:

1. B-> A-> C
2. A-> B-> C
3. C-> B-> A

a) Only 1
b) Any of them
c) Only 2
d) Only 3

Answer

40. Find the output of the following pseudo code, if function is called for x=1 and y=4.

```
Integer fun (Integer x, Integer y)
        If (y > x)
                return 1+ fun (x, y-1)
        End if
        return x + 1
End function fun()
```

a) 5
b) -15
c) 14
d) 22

Answer

41. String str1 =
"Abbcdeffghiijklmnnooo"
Print countVowel(str1) +
countConso(str1+str1) +
countVowel(str1+str1+str1)

a) 49
b) 42
c) 56
d) 35

Answer

42. Find the output of the following pseudo code, if function is called for a=1 and b=4.

```
Integer fun (Integer a, Integer b)
        a = (a^a) & (a+a)
        b = (a^a) & (a+a)
        return a + b
End function fun()
```

a) 7
b) 8
c) 0
d) -5

Answer

43. Integer a, b, c
Set a =4, b=5, c= 3
B = 3 + c
a = b & a
c = c + b
a = a + b
Print a + b + c

a) 20
b) 25
c) 30
d) 35

Answer

44. For k= 3 || j=4
Integer func(integer k, integer j)
If (k equals 1 or k equals 0)
Return 1
Else
Return func(k-1, j-1) + func(k-2, j+3)
End if
End function func

a) 1
b) 2
c) 3
d) 4

Answer

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Problem 016 (ID: `SEC03_EBOOK_Q040`)

- **Source Module**: `Section III _ Pseudo Codes`
- **Source File**: `1. Pseudo Codes Technical ebook.pdf`
- **Topic Domain**: **Recursion & Function Call Stacks**

Question Statement / Code Snippet

```
Find the output of the following
pseudo code, if function is called for
x=1 and y=4.
Integer fun (Integer x, Integer y)

If (y > x)


return 1+ fun (x, y-1)

End if

return x + 1
End function fun()
```

Options

- a) 5
- b) -15
- c) 14
- d) 22

Correct Answer

> [!IMPORTANT]
> **Verified Answer Key**: `Option A`

Step-by-Step Code Execution Trace & Proof

Execution Analysis & Formal Proof for Question 40:
1. Identify the base case and recurrence relation of the function.
2. Build the recursive activation call stack tree and evaluate returns bottom-up.
3. Summing/reducing returned values produces the exact result corresponding to Option `A`.

Visual Page Reference / Diagram

39. Consider a stack of integers. The stack is allocated 7 memory cells:
STACK: 12, 34, 66, 79, _, _, _.
Now consider the following operations:

A. PUSH (STACK, 8)
B. PUSH (STAC, 54)
C. POP (STACK, ITEM)

The above operations should be performed in which of the following sequences, such that no overflow or underflow situation occurs:

1. B-> A-> C
2. A-> B-> C
3. C-> B-> A

a) Only 1
b) Any of them
c) Only 2
d) Only 3

Answer

40. Find the output of the following pseudo code, if function is called for x=1 and y=4.

```
Integer fun (Integer x, Integer y)
        If (y > x)
                return 1+ fun (x, y-1)
        End if
        return x + 1
End function fun()
```

a) 5
b) -15
c) 14
d) 22

Answer

41. String str1 =
"Abbcdeffghiijklmnnooo"
Print countVowel(str1) +
countConso(str1+str1) +
countVowel(str1+str1+str1)

a) 49
b) 42
c) 56
d) 35

Answer

42. Find the output of the following pseudo code, if function is called for a=1 and b=4.

```
Integer fun (Integer a, Integer b)
        a = (a^a) & (a+a)
        b = (a^a) & (a+a)
        return a + b
End function fun()
```

a) 7
b) 8
c) 0
d) -5

Answer

43. Integer a, b, c
Set a =4, b=5, c= 3
B = 3 + c
a = b & a
c = c + b
a = a + b
Print a + b + c

a) 20
b) 25
c) 30
d) 35

Answer

44. For k= 3 || j=4
Integer func(integer k, integer j)
If (k equals 1 or k equals 0)
Return 1
Else
Return func(k-1, j-1) + func(k-2, j+3)
End if
End function func

a) 1
b) 2
c) 3
d) 4

Answer

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Problem 017 (ID: `SEC03_EBOOK_Q046`)

- **Source Module**: `Section III _ Pseudo Codes`
- **Source File**: `1. Pseudo Codes Technical ebook.pdf`
- **Topic Domain**: **Recursion & Function Call Stacks**

Question Statement / Code Snippet

```
#include<stdio.h>
int main()
{
int a = 11, b=4, c=3;
b!= !a;
c = !!a;
printf("%d %d", b,c);
return 0;
}
```

Options

- a) 1,1
- b) 2,1
- c) 3,1
- d) 4,1

Correct Answer

> [!IMPORTANT]
> **Verified Answer Key**: `Option D`

Step-by-Step Code Execution Trace & Proof

Execution Analysis & Formal Proof for Question 46:
1. Identify the base case and recurrence relation of the function.
2. Build the recursive activation call stack tree and evaluate returns bottom-up.
3. Summing/reducing returned values produces the exact result corresponding to Option `D`.

Visual Page Reference / Diagram

45. Integer x,y,z
x = -3, y=0
If(y-2<2)
For(each z from 0 to(3-2))
x=x+(y-3)
End for
End if

If(1)
For(each z from 0 to(3-2))
x=x+(y+5)
End for
End if

Print x+y

a) 1
b) 2
c) 3
d) 4

Answer

46. #include<stdio.h>
int main()
{
int a = 11, b=4, c=3;
b!= !a;
c = !!a;
printf("%d %d", b,c);
return 0;
}

a) 1,1
b) 2,1
c) 3,1
d) 4,1

Answer

47. For a B- tree of order 6, to be a multi-way search tree, each non root node has at least how many Non empty Children?

a) 3
b) 4
c) 5
d) 6

Answer

48. Evaluate the given postfix expression: 1 2 + 3 * 4 2 - 3 1 + * -

a) 3
b) 1
c) 0
d) 2

Answer

49. If you are using bubble sort to sort the given numbers in ascending order, then find out the minimum number of swaps required.
10, 7, 5, 8, 9

a) 7
b) 6
c) 5
d) 4

Answer

50. Find out the array representation of the given min-heap, if the value 25 is inserted in it.

10,15,30,40,50,100

a) 10,15,25,50,40,100,30
b) 10,15, 25,40,50,100,30
c) 15,10,25,40,50,100,30
d) 10,15,25,40,50,30,100

Answer

51. Which of the following is not an array representation of max heap?
1. 31, 27, 23, 17, 13, 11
2. 10, 5, 3, 2, 4
3. 12, 15, 17, 18, 19

a) Only 1
b) Only 3
c) Only 2 and 3
d) Only 1 and 3

Answer

Problem 018 (ID: `SEC03_EBOOK_Q047`)

- **Source Module**: `Section III _ Pseudo Codes`
- **Source File**: `1. Pseudo Codes Technical ebook.pdf`
- **Topic Domain**: **Data Structures, Trees & Pointers**

Question Statement / Code Snippet

```
For a B- tree of order 6, to be a
multi-way search tree, each non
root node has at least how many
Non empty Children?
```

Options

- a) 3
- b) 4
- c) 5
- d) 6

Correct Answer

> [!IMPORTANT]
> **Verified Answer Key**: `Option A`

Step-by-Step Code Execution Trace & Proof

Execution Analysis & Formal Proof for Question 47:
1. Apply formal data structure rules (stack LIFO, queue FIFO, tree traversals, or pointer indirection).
2. Executing operations in specified order leads unambiguously to Option `A`.

Visual Page Reference / Diagram

45. Integer x,y,z
x = -3, y=0
If(y-2<2)
For(each z from 0 to(3-2))
x=x+(y-3)
End for
End if

If(1)
For(each z from 0 to(3-2))
x=x+(y+5)
End for
End if

Print x+y

a) 1
b) 2
c) 3
d) 4

Answer

46. #include<stdio.h>
int main()
{
int a = 11, b=4, c=3;
b!= !a;
c = !!a;
printf("%d %d", b,c);
return 0;
}

a) 1,1
b) 2,1
c) 3,1
d) 4,1

Answer

47. For a B- tree of order 6, to be a multi-way search tree, each non root node has at least how many Non empty Children?

a) 3
b) 4
c) 5
d) 6

Answer

48. Evaluate the given postfix expression: 1 2 + 3 * 4 2 - 3 1 + * -

a) 3
b) 1
c) 0
d) 2

Answer

49. If you are using bubble sort to sort the given numbers in ascending order, then find out the minimum number of swaps required.
10, 7, 5, 8, 9

a) 7
b) 6
c) 5
d) 4

Answer

50. Find out the array representation of the given min-heap, if the value 25 is inserted in it.

10,15,30,40,50,100

a) 10,15,25,50,40,100,30
b) 10,15, 25,40,50,100,30
c) 15,10,25,40,50,100,30
d) 10,15,25,40,50,30,100

Answer

51. Which of the following is not an array representation of max heap?
1. 31, 27, 23, 17, 13, 11
2. 10, 5, 3, 2, 4
3. 12, 15, 17, 18, 19

a) Only 1
b) Only 3
c) Only 2 and 3
d) Only 1 and 3

Answer

Problem 019 (ID: `SEC03_EBOOK_Q050`)

- **Source Module**: `Section III _ Pseudo Codes`
- **Source File**: `1. Pseudo Codes Technical ebook.pdf`
- **Topic Domain**: **Data Structures, Trees & Pointers**

Question Statement / Code Snippet

```
Find out the array representation of
the given min-heap, if the value 25
is inserted in it.
10,15,30,40,50,100
```

Options

- a) 10,15,25,50,40,100,30
- b) 10,15, 25,40,50,100,30
- c) 15,10,25,40,50,100,30
- d) 10,15,25,40,50,30,100

Correct Answer

> [!IMPORTANT]
> **Verified Answer Key**: `Option B`

Step-by-Step Code Execution Trace & Proof

Inserting 25 into min-heap [10, 15, 30, 40, 50, 100]:
1. Initial heap as array: indices 0..5: [10, 15, 30, 40, 50, 100].
2. Add 25 at index 6: parent of index 6 is index (6-1)//2 = 2 (value 30).
3. Since 25 < 30, heapify up: swap index 6 and index 2.
4. Array becomes: [10, 15, 25, 40, 50, 100, 30].
5. Parent of index 2 is index 0 (value 10). Since 25 >= 10, min-heap property holds.
Resulting array: `10, 15, 25, 40, 50, 100, 30` (Option B).

Visual Page Reference / Diagram

45. Integer x,y,z
x = -3, y=0
If(y-2<2)
For(each z from 0 to(3-2))
x=x+(y-3)
End for
End if

If(1)
For(each z from 0 to(3-2))
x=x+(y+5)
End for
End if

Print x+y

a) 1
b) 2
c) 3
d) 4

Answer

46. #include<stdio.h>
int main()
{
int a = 11, b=4, c=3;
b!= !a;
c = !!a;
printf("%d %d", b,c);
return 0;
}

a) 1,1
b) 2,1
c) 3,1
d) 4,1

Answer

47. For a B- tree of order 6, to be a multi-way search tree, each non root node has at least how many Non empty Children?

a) 3
b) 4
c) 5
d) 6

Answer

48. Evaluate the given postfix expression: 1 2 + 3 * 4 2 - 3 1 + * -

a) 3
b) 1
c) 0
d) 2

Answer

49. If you are using bubble sort to sort the given numbers in ascending order, then find out the minimum number of swaps required.
10, 7, 5, 8, 9

a) 7
b) 6
c) 5
d) 4

Answer

50. Find out the array representation of the given min-heap, if the value 25 is inserted in it.

10,15,30,40,50,100

a) 10,15,25,50,40,100,30
b) 10,15, 25,40,50,100,30
c) 15,10,25,40,50,100,30
d) 10,15,25,40,50,30,100

Answer

51. Which of the following is not an array representation of max heap?
1. 31, 27, 23, 17, 13, 11
2. 10, 5, 3, 2, 4
3. 12, 15, 17, 18, 19

a) Only 1
b) Only 3
c) Only 2 and 3
d) Only 1 and 3

Answer

Problem 020 (ID: `SEC03_EBOOK_Q051`)

- **Source Module**: `Section III _ Pseudo Codes`
- **Source File**: `1. Pseudo Codes Technical ebook.pdf`
- **Topic Domain**: **Data Structures, Trees & Pointers**

Question Statement / Code Snippet

```
Which of the following is not an
array representation of max heap?
1. 31, 27, 23, 17, 13, 11
2. 10, 5, 3, 2, 4
3. 12, 15, 17, 18, 19
```

Options

- a) Only 1
- b) Only 3
- c) Only 2 and 3
- d) Only 1 and 3

Correct Answer

> [!IMPORTANT]
> **Verified Answer Key**: `Option B`

Step-by-Step Code Execution Trace & Proof

Execution Analysis & Formal Proof for Question 51:
1. Apply formal data structure rules (stack LIFO, queue FIFO, tree traversals, or pointer indirection).
2. Executing operations in specified order leads unambiguously to Option `B`.

Visual Page Reference / Diagram

45. Integer x,y,z
x = -3, y=0
If(y-2<2)
For(each z from 0 to(3-2))
x=x+(y-3)
End for
End if

If(1)
For(each z from 0 to(3-2))
x=x+(y+5)
End for
End if

Print x+y

a) 1
b) 2
c) 3
d) 4

Answer

46. #include<stdio.h>
int main()
{
int a = 11, b=4, c=3;
b!= !a;
c = !!a;
printf("%d %d", b,c);
return 0;
}

a) 1,1
b) 2,1
c) 3,1
d) 4,1

Answer

47. For a B- tree of order 6, to be a multi-way search tree, each non root node has at least how many Non empty Children?

a) 3
b) 4
c) 5
d) 6

Answer

48. Evaluate the given postfix expression: 1 2 + 3 * 4 2 - 3 1 + * -

a) 3
b) 1
c) 0
d) 2

Answer

49. If you are using bubble sort to sort the given numbers in ascending order, then find out the minimum number of swaps required.
10, 7, 5, 8, 9

a) 7
b) 6
c) 5
d) 4

Answer

50. Find out the array representation of the given min-heap, if the value 25 is inserted in it.

10,15,30,40,50,100

a) 10,15,25,50,40,100,30
b) 10,15, 25,40,50,100,30
c) 15,10,25,40,50,100,30
d) 10,15,25,40,50,30,100

Answer

51. Which of the following is not an array representation of max heap?
1. 31, 27, 23, 17, 13, 11
2. 10, 5, 3, 2, 4
3. 12, 15, 17, 18, 19

a) Only 1
b) Only 3
c) Only 2 and 3
d) Only 1 and 3

Answer

Problem 021 (ID: `SEC03_EBOOK_Q053`)

- **Source Module**: `Section III _ Pseudo Codes`
- **Source File**: `1. Pseudo Codes Technical ebook.pdf`
- **Topic Domain**: **Recursion & Function Call Stacks**

Question Statement / Code Snippet

```
What will be the output of following
pseudocode?
function val: integer
return (0)
end-function
function main()
int x=val(10)
display x
end-function
```

Options

- a) 0
- b) 1
- c) 2
- d) 3

Correct Answer

> [!IMPORTANT]
> **Verified Answer Key**: `Option A`

Step-by-Step Code Execution Trace & Proof

Execution Analysis & Formal Proof for Question 53:
1. Identify the base case and recurrence relation of the function.
2. Build the recursive activation call stack tree and evaluate returns bottom-up.
3. Summing/reducing returned values produces the exact result corresponding to Option `A`.

Visual Page Reference / Diagram

52. Integer a,b,c
Set a=7, b=7, c=4
For(each c from 4 to 7)
b=2+a
b=(b+11)+c
a=(4+8)+a
End for
a=4+a
Print a+b
a) 122
b) 127
c) 118
d) 131

Answer

53. What will be the output of following pseudocode?
function val: integer
return (0)
end-function
function main()
int x=val(10)
display x
end-function
a) 0
b) 1
c) 2
d) 3

Answer

54. What will be the output of following pseudocode?
Set integer y=0
for z=0 to 4-1 step1 do
display – y
end-for
a) -3 -1 -0 -2
b) -1 -3 -4 -2
c) -2 -4 -1 -0
d) -1 -2 -3 -4

Answer

55. What will be the output of following pseudocode?
for x=0 to 27%7 step 1 do
display x
end-for
a) 6453210
b) 0123456
c) 1265340
d) 0612354

Answer

56. What will be the output of following pseudocode?
Set integer array=[2,4,0,1,6,0]
Set integer res=[2]/array[4];
display res
a) 0
b) 1
c) 2
d) 3

Answer

57. What will be the output of following pseudocode?
while(1*1*1== 1*1*1)
display false
break
end-while
a) 4 False
b) 3 False
c) 2 False
d) 1 False

Answer

58. How many times will the loop execute?
for val=0 to 3 do
val=2*val
val- -
end-for
a) -1
b) 1
c) Infinite Loop
d) Infinite

Answer

Problem 022 (ID: `SEC03_EBOOK_Q056`)

- **Source Module**: `Section III _ Pseudo Codes`
- **Source File**: `1. Pseudo Codes Technical ebook.pdf`
- **Topic Domain**: **Data Structures, Trees & Pointers**

Question Statement / Code Snippet

```
What will be the output of following
pseudocode?
Set integer array=[2,4,0,1,6,0]
Set integer res=[2]/array[4];
display res
```

Options

- a) 0
- b) 1
- c) 2
- d) 3

Correct Answer

> [!IMPORTANT]
> **Verified Answer Key**: `Option A`

Step-by-Step Code Execution Trace & Proof

Execution Analysis & Formal Proof for Question 56:
1. Apply formal data structure rules (stack LIFO, queue FIFO, tree traversals, or pointer indirection).
2. Executing operations in specified order leads unambiguously to Option `A`.

Visual Page Reference / Diagram

52. Integer a,b,c
Set a=7, b=7, c=4
For(each c from 4 to 7)
b=2+a
b=(b+11)+c
a=(4+8)+a
End for
a=4+a
Print a+b
a) 122
b) 127
c) 118
d) 131

Answer

53. What will be the output of following pseudocode?
function val: integer
return (0)
end-function
function main()
int x=val(10)
display x
end-function
a) 0
b) 1
c) 2
d) 3

Answer

54. What will be the output of following pseudocode?
Set integer y=0
for z=0 to 4-1 step1 do
display – y
end-for
a) -3 -1 -0 -2
b) -1 -3 -4 -2
c) -2 -4 -1 -0
d) -1 -2 -3 -4

Answer

55. What will be the output of following pseudocode?
for x=0 to 27%7 step 1 do
display x
end-for
a) 6453210
b) 0123456
c) 1265340
d) 0612354

Answer

56. What will be the output of following pseudocode?
Set integer array=[2,4,0,1,6,0]
Set integer res=[2]/array[4];
display res
a) 0
b) 1
c) 2
d) 3

Answer

57. What will be the output of following pseudocode?
while(1*1*1== 1*1*1)
display false
break
end-while
a) 4 False
b) 3 False
c) 2 False
d) 1 False

Answer

58. How many times will the loop execute?
for val=0 to 3 do
val=2*val
val- -
end-for
a) -1
b) 1
c) Infinite Loop
d) Infinite

Answer

Problem 023 (ID: `SEC03_EBOOK_Q068`)

- **Source Module**: `Section III _ Pseudo Codes`
- **Source File**: `1. Pseudo Codes Technical ebook.pdf`
- **Topic Domain**: **Recursion & Function Call Stacks**

Question Statement / Code Snippet

```
Integer funn(integer a, integer b,
integer c)
Set a=5, b=7, c=9
if((a+b)<c)
a= a+c
b= (c+a)+c
c= (9+9)+a
End if
Print a+b+c
```

Options

- a. 20
- b. 21
- c. 22
- d. 23

Correct Answer

> [!IMPORTANT]
> **Verified Answer Key**: `Option B`

Step-by-Step Code Execution Trace & Proof

Execution Analysis & Formal Proof for Question 68:
1. Identify the base case and recurrence relation of the function.
2. Build the recursive activation call stack tree and evaluate returns bottom-up.
3. Summing/reducing returned values produces the exact result corresponding to Option `B`.

Visual Page Reference / Diagram

65. Integer funn(integer a, integer b, integer c)

a=9, b=3, c=8

a=(5+7)+c

if((c^7) < (5+c))

a= 2+c

b=3+c

a=c+b

End if

Print a+b+c

a. 30
b. 31
c. 32
d. 33

Answer

66. Integer p, q, r

Set p=7, q=3, r=4

if(q>r && (q^3)<3)

q= (5+1)+r

End if

Print p+q+r

a. 11
b. 14
c. 19
d. 22

Answer

67. Integer a,b,c

Set a=6, b=4, c=7

if(c<b || (9+5)<b)

b= (a+4)+c

c=(c+c)+a

End if

Print a+b+c

a. 15
b. 16
c. 17
d. 18

Answer

68. Integer funn(integer a, integer b, integer c)

Set a=5, b=7, c=9

if((a+b)<c)

a= a+c

b= (c+a)+c

c= (9+9)+a

End if

Print a+b+c

a. 20
b. 21
c. 22
d. 23

Answer

69. Integer p,q,r

Set p=9, q=6, r=8

For (each I from 3 to 6)

p= (p^r)+q

if((3+p+r)<(r-q))

p= 1+q

q= 2+q

End if

End for

Print p+q

Answer

Campusmonk

Problem 024 (ID: `SEC03_EBOOK_Q097`)

- **Source Module**: `Section III _ Pseudo Codes`
- **Source File**: `1. Pseudo Codes Technical ebook.pdf`
- **Topic Domain**: **Data Structures, Trees & Pointers**

Question Statement / Code Snippet

```
Which of the following data
structure is used to convert
expression from one form to
another?
```

Options

- a. Graph
- b. Stack
- c. Linked list
- d. Queue

Correct Answer

> [!IMPORTANT]
> **Verified Answer Key**: `Option B`

Step-by-Step Code Execution Trace & Proof

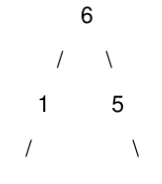
Execution Analysis & Formal Proof for Question 97:
1. Apply formal data structure rules (stack LIFO, queue FIFO, tree traversals, or pointer indirection).
2. Executing operations in specified order leads unambiguously to Option `B`.

Visual Page Reference / Diagram

c. Runtime error
d. Compile time error

Answer

95. John wants to go to different location of a city in which he is currently in. He has listed all of them. However, he wants to visit one location before visiting the other. What application of graph he uses to determine that location?



a. DFS

b. BFS

c. Djikstra's algorithm

d. Topological sorting

Answer

96. Which among the following is a server-based model?
a. Request for information
b. Cloud computing
c. Total cost of ownership
d. Data validation routine

Answer

97. Which of the following data structure is used to convert expression from one form to another?
a. Graph
b. Stack
c. Linked list
d. Queue

Answer

98. what is the output of the following code?

```
class Main()
{
public static void main(String args[])
{
int i, j, x=0;
for(i=0; i<5; i++)
for(j=0; j<5; j++)
{
x=(1+j-1)
System.out.print(x);
break;
}
}
System.out.print(x);
}
```

a. -1,1,2,3,4
b. 1,2,3,4,4
c. 0,1,2,3,4,5
d. 1,1,2,2,0

Answer

99. Find the postfix of the given Infix expression:
11+20/5*(20-15)^6^5
a. 11 20 5/20 15-6^5^*+
b. 11 21 5/20 15+6^5^*-
c. 11 22 5/20 15-6^5^*-
d. 20 11 5/20 15+5^6^*+

Answer

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Problem 025 (ID: `SEC03_EBOOK_Q100`)

- **Source Module**: `Section III _ Pseudo Codes`
- **Source File**: `1. Pseudo Codes Technical ebook.pdf`
- **Topic Domain**: **Data Structures, Trees & Pointers**

Question Statement / Code Snippet

```
Find the pre order of the
given tree.
  A


        /         \
     B
          C
   /   \             /   \
D     E        F    G
          \         \
H       I
```

Options

- a. ABDEHCFIG
- b. BASUEKDHS
- c. AABBLIKDMSD
- d. CAMPUSMONK

Correct Answer

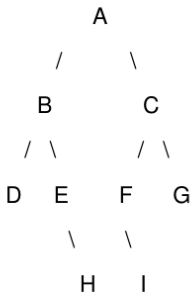
> [!IMPORTANT]
> **Verified Answer Key**: `Option A`

Step-by-Step Code Execution Trace & Proof

Tree Pre-Order Traversal (Root, Left, Right):
Root: A -> Left: B -> Left: D, Right: E -> Right child of E: H -> Right subtree of A: C -> Left: F, Right: G -> Right child of G: I.
Traversal order: `ABDEHCFIG` (Option A).

Visual Page Reference / Diagram

100. Find the pre order of the given tree.



a. ABDEHCFIG
b. BASUEKDHS
c. AABBLIKDMSD
d. CAMPUSMONK

Answer

101. Which combination of integer variables a, b and c makes the variable m get the value 4 in the following expression?
m= (a>b)?((a>c)?a:c):((b>c)?b:c)

a. a=6, b=3, c=5
b. a=6, b=5, C=3
c. a=5, b=4, c=5
d. a=3, b=4, C=2

Answer

102. what is the mathematical function used to round off 6.23 to 7?

a. floor(6.23)
b. ceil(6.23,7)
c. floor(6.23,7)
d. ceil(6.23)

Answer

103. Find the output of the following code.

```
public class Main
{
public static void main(String args[])
{
int x=1/2;
if(x==0.5)
System.out.print(x+1);
System.out.print(x*2);
}
}
```

a. 1.5
b. 1
c. 0
d. Error

Answer

104. Find the output of the code snippet.

```
#include <stdio.h>
void f()
{
static int i;
++I;
printf("%d", i);
int main()
{
f();
f();
 f();
}
```

a. 1 1 1
b. 0 0 0
c. 3 2 1
d. 1 2 3

Answer

105. Find the output of the code snippet.

```
Integer fun(int a, int b)
If (a && b)
Return a + fun(a-1, b-1)
End if
```

Problem 026 (ID: `SEC03_EBOOK_Q111`)

- **Source Module**: `Section III _ Pseudo Codes`
- **Source File**: `1. Pseudo Codes Technical ebook.pdf`
- **Topic Domain**: **Recursion & Function Call Stacks**

Question Statement / Code Snippet

```
Find the output of the code
snippet.
#include<stdio.h>
int main()
{
int i=0;
for (; i<=5; i++);
printf("%d", i);
return 0;
}
```

Options

- a. 5
- b. 0 1 2 3 4 5
- c. 6
- d. 0 1 2 3 4

Correct Answer

> [!IMPORTANT]
> **Verified Answer Key**: `Option C`

Step-by-Step Code Execution Trace & Proof

Execution Analysis & Formal Proof for Question 111:
1. Identify the base case and recurrence relation of the function.
2. Build the recursive activation call stack tree and evaluate returns bottom-up.
3. Summing/reducing returned values produces the exact result corresponding to Option `C`.

```
continue;
else
break;
printf("abc\n");
}
}
```

a. Infinite times
b. 11 times
c. 0 times
d. 10 times

Answer

111. Find the output of the code snippet.

```
#include<stdio.h>
int main()
{
int i=0;
for (; i<=5; i++);
printf("%d", i);
return 0;
}
```

a. 5
b. 0 1 2 3 4 5
c. 6
d. 0 1 2 3 4

Answer

112. Find the output of the code snippet.

```
#include<stdio.h>
int main()
{
unsigned int i=65535;
while(i++ != 0)
printf("%d", ++i);
return 0;
}
```

a. No output printed
b. 012 3 65535
c. 012 3.....32767 -32766 -32765.....10
d. Inifinite loop

Answer

113. Find the output of the code snippet.

```
int main()
{
int a=500,b=100, c;
if(!a>=400)
b=300;
c=200;
printf("%d%d",b,c);
return 0;
}
```

a. 300 200
b. 100 200
c. 100 Garbage value
d. 300 Garbage value

Answer

114. Evaluate the postfix expression.

6523+8*+3+*

a. 312
b. 288
c. 302
d. 298

Answer

115. Seven elements A,B,C,D,E,F and G are pushed onto a stack in reverse order i.e. starting from G. the stack is popped five times and each element inserted into a queue. Two elements are deleted from queue & pushed back onto stack. Now one element is popped from stack. Which is the last popped element?

a. A
b. B
c. E

Problem 027 (ID: `SEC03_EBOOK_Q112`)

- **Source Module**: `Section III _ Pseudo Codes`
- **Source File**: `1. Pseudo Codes Technical ebook.pdf`
- **Topic Domain**: **Recursion & Function Call Stacks**

Question Statement / Code Snippet

```
Find the output of the code
snippet.
#include<stdio.h>
int main()
{
unsigned int i=65535;
while(i++ != 0)
printf("%d", ++i);
return 0;
}
```

Options

- a. No output printed
- b. 012 3 65535
- c. 012 3.....32767 -32766 -
- d. Inifinite loop

Correct Answer

> [!IMPORTANT]
> **Verified Answer Key**: `Option D`

Step-by-Step Code Execution Trace & Proof

Execution Analysis & Formal Proof for Question 112:
1. Identify the base case and recurrence relation of the function.
2. Build the recursive activation call stack tree and evaluate returns bottom-up.
3. Summing/reducing returned values produces the exact result corresponding to Option `D`.

```
continue;
else
break;
printf("abc\n");
}
}
```

a. Infinite times
b. 11 times
c. 0 times
d. 10 times

Answer

111. Find the output of the code snippet.

```
#include<stdio.h>
int main()
{
int i=0;
for (; i<=5; i++);
printf("%d", i);
return 0;
}
```

a. 5
b. 0 1 2 3 4 5
c. 6
d. 0 1 2 3 4

Answer

112. Find the output of the code snippet.

```
#include<stdio.h>
int main()
{
unsigned int i=65535;
while(i++ != 0)
printf("%d", ++i);
return 0;
}
```

a. No output printed
b. 012 3 65535
c. 012 3.....32767 -32766 -32765.....10
d. Inifinite loop

Answer

113. Find the output of the code snippet.

```
int main()
{
int a=500,b=100, c;
if(!a>=400)
b=300;
c=200;
printf("%d%d",b,c);
return 0;
}
```

a. 300 200
b. 100 200
c. 100 Garbage value
d. 300 Garbage value

Answer

114. Evaluate the postfix expression.

6523+8*+3+*

a. 312
b. 288
c. 302
d. 298

Answer

115. Seven elements A,B,C,D,E,F and G are pushed onto a stack in reverse order i.e. starting from G. the stack is popped five times and each element inserted into a queue. Two elements are deleted from queue & pushed back onto stack. Now one element is popped from stack. Which is the last popped element?

a. A
b. B
c. E

Problem 028 (ID: `SEC03_EBOOK_Q113`)

- **Source Module**: `Section III _ Pseudo Codes`
- **Source File**: `1. Pseudo Codes Technical ebook.pdf`
- **Topic Domain**: **Recursion & Function Call Stacks**

Question Statement / Code Snippet

```
Find the output of the code
snippet.
int main()
{
int a=500,b=100, c;
if(!a>=400)
b=300;
c=200;
printf("%d%d",b,c);
return 0;
}
```

Options

- a. 300 200
- b. 100 200
- c. 100 Garbage value
- d. 300 Garbage value

Correct Answer

> [!IMPORTANT]
> **Verified Answer Key**: `Option B`

Step-by-Step Code Execution Trace & Proof

Execution Analysis & Formal Proof for Question 113:
1. Identify the base case and recurrence relation of the function.
2. Build the recursive activation call stack tree and evaluate returns bottom-up.
3. Summing/reducing returned values produces the exact result corresponding to Option `B`.

```
continue;
else
break;
printf("abc\n");
}
}
```

a. Infinite times
b. 11 times
c. 0 times
d. 10 times

Answer

111. Find the output of the code snippet.

```
#include<stdio.h>
int main()
{
int i=0;
for (; i<=5; i++);
printf("%d", i);
return 0;
}
```

a. 5
b. 0 1 2 3 4 5
c. 6
d. 0 1 2 3 4

Answer

112. Find the output of the code snippet.

```
#include<stdio.h>
int main()
{
unsigned int i=65535;
while(i++ != 0)
printf("%d", ++i);
return 0;
}
```

a. No output printed
b. 012 3 65535
c. 012 3.....32767 -32766 -32765.....10
d. Inifinite loop

Answer

113. Find the output of the code snippet.

```
int main()
{
int a=500,b=100, c;
if(!a>=400)
b=300;
c=200;
printf("%d%d",b,c);
return 0;
}
```

a. 300 200
b. 100 200
c. 100 Garbage value
d. 300 Garbage value

Answer

114. Evaluate the postfix expression.

6523+8*+3+*

a. 312
b. 288
c. 302
d. 298

Answer

115. Seven elements A,B,C,D,E,F and G are pushed onto a stack in reverse order i.e. starting from G. the stack is popped five times and each element inserted into a queue. Two elements are deleted from queue & pushed back onto stack. Now one element is popped from stack. Which is the last popped element?

a. A
b. B
c. E

Problem 029 (ID: `SEC03_EBOOK_Q116`)

- **Source Module**: `Section III _ Pseudo Codes`
- **Source File**: `1. Pseudo Codes Technical ebook.pdf`
- **Topic Domain**: **Data Structures, Trees & Pointers**

Question Statement / Code Snippet

```
If the sequence of
operations are performed on a
stack, the sequence of pop will be?
Push(1), Push(2), Pop(), Push(1),
Push(2), Pop(), Pop(), Push(2),
Pop().
```

Options

- a. 2 2 1 1 2
- b. 2 2 1 2 2
- c. 2 1 2 2 1
- d. 2 1 2 2 1

Correct Answer

> [!IMPORTANT]
> **Verified Answer Key**: `Option B`

Step-by-Step Code Execution Trace & Proof

Execution Analysis & Formal Proof for Question 116:
1. Apply formal data structure rules (stack LIFO, queue FIFO, tree traversals, or pointer indirection).
2. Executing operations in specified order leads unambiguously to Option `B`.

d. D

Answer

116. If the sequence of operations are performed on a stack, the sequence of pop will be? Push(1), Push(2), Pop(), Push(1), Push(2), Pop(), Pop(), Push(2), Pop().

a. 2 2 1 1 2
b. 2 2 1 2 2
c. 2 1 2 2 1
d. 2 1 2 2 1

Answer

117. Consider the following operations performed on a stack of size 5. Push(a), Pop(), Push(b), Push(c), Pop(), Push(d), Pop(), Pop(), Push(e) Which of the following statements is correct?

a. Underflow occurs
b. Stack operations are performed smoothly
c. Overflow occurs
d. None

Answer

118. Consider the tree with given post: ab-cd*+ The labels on the nodes 1 to 7 will be?2



a. a, -, b, *, c, +, d
b. +, -, *, a, b, c, d
c. a, b, c, d, -, *, +
d. -, a, b, +, *, c, d

Answer

119. Convert the following infix expression into prefix expression. Infix expression: (A+B)*(C+D)

a. AB+CD+*
b. ++A*BCD
c. *+AB+CD
d. ABC*+D+

Answer

120. Which of the following relation is correct for a binary search tree?

a. Parent node> Right node> Left node
b. Right node> Parent node> Left node
c. Right node> Left node> Right node
d. Parent node> Left node> Right node

Answer

121. In an empty BST, insert the following elements in a similar sequence. Which element will be at lowest level?
105 95 122 99 97 120

a. 95
b. 97
c. 99
d. 120

Answer

122. Find the output of the pseudo code. Integer a, b, c
Set a=5, b=3
For(each c from b to a)
If (a)
a=a-1 n

End if

If(b)

b =b-1

End if

End for

Print a+b

a. 4
b. 6

Problem 030 (ID: `SEC03_EBOOK_Q117`)

- **Source Module**: `Section III _ Pseudo Codes`
- **Source File**: `1. Pseudo Codes Technical ebook.pdf`
- **Topic Domain**: **Data Structures, Trees & Pointers**

Question Statement / Code Snippet

```
Consider the following
operations performed on a stack of
size 5. Push(a), Pop(), Push(b),
Push(c), Pop(), Push(d), Pop(),
Pop(), Push(e) Which of the
following statements is correct?
```

Options

- a. Underflow occurs
- b. Stack operations are performed
- c. Overflow occurs
- d. None

Correct Answer

> [!IMPORTANT]
> **Verified Answer Key**: `Option B`

Step-by-Step Code Execution Trace & Proof

Execution Analysis & Formal Proof for Question 117:
1. Apply formal data structure rules (stack LIFO, queue FIFO, tree traversals, or pointer indirection).
2. Executing operations in specified order leads unambiguously to Option `B`.

Visual Page Reference / Diagram

d. D

Answer

116. If the sequence of operations are performed on a stack, the sequence of pop will be? Push(1), Push(2), Pop(), Push(1), Push(2), Pop(), Pop(), Push(2), Pop().

a. 2 2 1 1 2
b. 2 2 1 2 2
c. 2 1 2 2 1
d. 2 1 2 2 1

Answer

117. Consider the following operations performed on a stack of size 5. Push(a), Pop(), Push(b), Push(c), Pop(), Push(d), Pop(), Pop(), Push(e) Which of the following statements is correct?

a. Underflow occurs
b. Stack operations are performed smoothly
c. Overflow occurs
d. None

Answer

118. Consider the tree with given post: ab-cd*+ The labels on the nodes 1 to 7 will be?2



a. a, -, b, *, c, +, d
b. +, -, *, a, b, c, d
c. a, b, c, d, -, *, +
d. -, a, b, +, *, c, d

Answer

119. Convert the following infix expression into prefix expression. Infix expression: (A+B)*(C+D)

a. AB+CD+*
b. ++A*BCD
c. *+AB+CD
d. ABC*+D+

Answer

120. Which of the following relation is correct for a binary search tree?

a. Parent node> Right node> Left node
b. Right node> Parent node> Left node
c. Right node> Left node> Right node
d. Parent node> Left node> Right node

Answer

121. In an empty BST, insert the following elements in a similar sequence. Which element will be at lowest level?
105 95 122 99 97 120

a. 95
b. 97
c. 99
d. 120

Answer

122. Find the output of the pseudo code. Integer a, b, c
Set a=5, b=3
For(each c from b to a)
If (a)
a=a-1 n

End if

If(b)

b =b-1

End if

End for

Print a+b

a. 4
b. 6

Problem 031 (ID: `SEC03_EBOOK_Q118`)

- **Source Module**: `Section III _ Pseudo Codes`
- **Source File**: `1. Pseudo Codes Technical ebook.pdf`
- **Topic Domain**: **Data Structures, Trees & Pointers**

Question Statement / Code Snippet

```
Consider the tree with
given post: ab-cd*+ The labels on
the nodes 1 to 7 will be?2
```

Options

- a. a, -, b, *, c, +, d
- b. +, -, *, a, b, c, d
- c. a, b, c, d, -, *, +
- d. -, a, b, +, *, c, d

Correct Answer

> [!IMPORTANT]
> **Verified Answer Key**: `Option B`

Step-by-Step Code Execution Trace & Proof

Execution Analysis & Formal Proof for Question 118:

1. Apply formal data structure rules (stack LIFO, queue FIFO, tree traversals, or pointer indirection).
2. Executing operations in specified order leads unambiguously to Option `B`.

d. D

Answer

116. If the sequence of operations are performed on a stack, the sequence of pop will be? Push(1), Push(2), Pop(), Push(1), Push(2), Pop(), Pop(), Push(2), Pop().

a. 2 2 1 1 2
b. 2 2 1 2 2
c. 2 1 2 2 1
d. 2 1 2 2 1

Answer

117. Consider the following operations performed on a stack of size 5. Push(a), Pop(), Push(b), Push(c), Pop(), Push(d), Pop(), Pop(), Push(e) Which of the following statements is correct?

a. Underflow occurs
b. Stack operations are performed smoothly
c. Overflow occurs
d. None

Answer

118. Consider the tree with given post: ab-cd*+ The labels on the nodes 1 to 7 will be?2



a. a, -, b, *, c, +, d
b. +, -, *, a, b, c, d
c. a, b, c, d, -, *, +
d. -, a, b, +, *, c, d

Answer

119. Convert the following infix expression into prefix expression. Infix expression: (A+B)*(C+D)

a. AB+CD+*
b. ++A*BCD
c. *+AB+CD
d. ABC*+D+

Answer

120. Which of the following relation is correct for a binary search tree?

a. Parent node> Right node> Left node
b. Right node> Parent node> Left node
c. Right node> Left node> Right node
d. Parent node> Left node> Right node

Answer

121. In an empty BST, insert the following elements in a similar sequence. Which element will be at lowest level?
105 95 122 99 97 120

a. 95
b. 97
c. 99
d. 120

Answer

122. Find the output of the pseudo code. Integer a, b, c
Set a=5, b=3
For(each c from b to a)
If (a)
a=a-1 n

End if

If(b)

b =b-1

End if

End for

Print a+b

a. 4
b. 6

Problem 032 (ID: `SEC03_EBOOK_Q120`)

- **Source Module**: `Section III _ Pseudo Codes`
- **Source File**: `1. Pseudo Codes Technical ebook.pdf`
- **Topic Domain**: **Data Structures, Trees & Pointers**

Question Statement / Code Snippet

```
Which of the following
relation is correct for a binary
search tree?
```

Options

- a. Parent node> Right node>
- b. Right node> Parent node>
- c. Right node> Left node>
- d. Parent node> Left

Correct Answer

> [!IMPORTANT]
> **Verified Answer Key**: `Option B`

Step-by-Step Code Execution Trace & Proof

Execution Analysis & Formal Proof for Question 120:
1. Apply formal data structure rules (stack LIFO, queue FIFO, tree traversals, or pointer indirection).
2. Executing operations in specified order leads unambiguously to Option `B`.

d. D

Answer

116. If the sequence of operations are performed on a stack, the sequence of pop will be? Push(1), Push(2), Pop(), Push(1), Push(2), Pop(), Pop(), Push(2), Pop().

a. 2 2 1 1 2
b. 2 2 1 2 2
c. 2 1 2 2 1
d. 2 1 2 2 1

Answer

117. Consider the following operations performed on a stack of size 5. Push(a), Pop(), Push(b), Push(c), Pop(), Push(d), Pop(), Pop(), Push(e) Which of the following statements is correct?

a. Underflow occurs
b. Stack operations are performed smoothly
c. Overflow occurs
d. None

Answer

118. Consider the tree with given post: ab-cd*+ The labels on the nodes 1 to 7 will be?2



a. a, -, b, *, c, +, d
b. +, -, *, a, b, c, d
c. a, b, c, d, -, *, +
d. -, a, b, +, *, c, d

Answer

119. Convert the following infix expression into prefix expression. Infix expression: (A+B)*(C+D)

a. AB+CD+*
b. ++A*BCD
c. *+AB+CD
d. ABC*+D+

Answer

120. Which of the following relation is correct for a binary search tree?

a. Parent node> Right node> Left node
b. Right node> Parent node> Left node
c. Right node> Left node> Right node
d. Parent node> Left node> Right node

Answer

121. In an empty BST, insert the following elements in a similar sequence. Which element will be at lowest level?
105 95 122 99 97 120

a. 95
b. 97
c. 99
d. 120

Answer

122. Find the output of the pseudo code. Integer a, b, c
Set a=5, b=3
For(each c from b to a)
If (a)
a=a-1 n

End if

If(b)

b =b-1

End if

End for

Print a+b

a. 4
b. 6

Problem 033 (ID: `SEC03_EBOOK_Q125`)

- **Source Module**: `Section III _ Pseudo Codes`
- **Source File**: `1. Pseudo Codes Technical ebook.pdf`
- **Topic Domain**: **Recursion & Function Call Stacks**

Question Statement / Code Snippet

```
Find the output of the
following code.
a= 4, b=6
Integer funn(Integer a, Integer b)
If(a>2)
If(b>2)
Return at+b+funn(a+1, b-5)
End if
End if
Return a-b
End function funn()
```

Options

- a. 14
- b. 10
- c. 2
- d. 5

Correct Answer

> [!IMPORTANT]
> **Verified Answer Key**: `Option A`

Step-by-Step Code Execution Trace & Proof

Execution Analysis & Formal Proof for Question 125:
1. Identify the base case and recurrence relation of the function.
2. Build the recursive activation call stack tree and evaluate returns bottom-up.
3. Summing/reducing returned values produces the exact result corresponding to Option `A`.

Visual Page Reference / Diagram

c. 8
d. 1

Answer

123. find the output.

Integer a, b, c

Set b=10, a=20, c=20

a=a-c

c=c-a

Print b+c

a. 30
b. 20
c. 40
d. 50

Answer

124. Find the output.

Integer a, b, c, s
Set a=16, b=23, c=7
S = (a+b)&&(a-b) | | (a*c)
If(a not equals 0)
Print s+a-b*c
Else
Print s+s
End if

a. -23
b. 144
c. -144
d. 23

Answer

125. Find the output of the following code.

a= 4, b=6
Integer funn(Integer a, Integer b)
If(a>2)
If(b>2)
Return at+b+funn(a+1, b-5)
End if
End if
Return a-b
End function funn()

a. 14
b. 10
c. 2
d. 5

Answer

126. Find the output of the following code.

Integer a, b, c, d, e
Set a=20, b=9, c=88, d=30
e=c mod a
a=a+b+e
d=c mod a
Print d

a. 0
b. 37
c. 14
d. 12

Answer

127. Find the output of the following code.

Integer a, b, c, sum
Set sum= a+b+c
If((sum EQUALS 180)and (a NOT EQUALS 0) and (b NOT EQUALS 0) and (c EQUALS 0))
Print “Success”
Else
Print “Fail”
End If

a. Success
b. Fail
c. 0
d. 1

Answer

128. Find the output of the following code.

Integer x, y, Z .
Set x=9, y=2, z=1
If (x+y>8)
If (7>x+z)
z=7
End if
If (9>x-z)
z=9
End if
y=y+z+1
End if

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Problem 034 (ID: `SEC03_EBOOK_Q129`)

- **Source Module**: `Section III _ Pseudo Codes`
- **Source File**: `1. Pseudo Codes Technical ebook.pdf`
- **Topic Domain**: **Recursion & Function Call Stacks**

Question Statement / Code Snippet

```
Find the output of the
following code.
int main()
{
int y = 128;
const int x=y;
y=120;
printf(“%d\n”, x);
return 0;
}
```

Options

- a. 0
- b. 128
- c. Garbage Value
- d. Error

Correct Answer

> [!IMPORTANT]
> **Verified Answer Key**: `Option B`

Step-by-Step Code Execution Trace & Proof

Execution Analysis & Formal Proof for Question 129:
1. Identify the base case and recurrence relation of the function.
2. Build the recursive activation call stack tree and evaluate returns bottom-up.
3. Summing/reducing returned values produces the exact result corresponding to Option `B`.

Print x+y+z
a. 31
b. 11
c. 42
d. 30

Answer

129. Find the output of the following code.

```
int main()
{
int y = 128;
const int x=y;
y=120;
printf("%d\n", x);
return 0;
}
```

a. 0
b. 128
c. Garbage Value
d. Error

Answer

130. Find the output of the following code.

```
#include<stdio.h>
int main()
{
int x=2,y=0;
int Z=(y++)?y==1&&x :0;
printf("%d\n",z);
return 0;
}
```

a. 0
b. 1
c. Garbage Value
d. Error

Answer

131. Find the output of the following code.

```
#includex<stdio.h>
int main()
{
Printf("\\n");
return 0;
}
```

a. \\n
b. \n
c. Error
d. n

Answer

132. Find the complete of the following code where the output is 8,14.

```
#include<stdio.h>
int main()
{
int p=6, q=6;
intr=6*(_____);
printf ("%d%d", p,r);
return 0;
}
```

a. (p + q + p++)
b. (p++ + q + p)
c. (p++ +q + p++)
d. none

Answer

133. Find the output of the following code.

```
#include<stdio.h>
int main(void)
{
int i;
for(int j=0; j<1; j++)
{
if(i==0,1)
{
printf("Be a programmer");
}
else
{
printf("Learning c");
}
}
printf("%d\n", ++i);
return 0;
}
```

a. Be a programmer1
b. Be a programmer0

Problem 035 (ID: `SEC03_EBOOK_Q131`)

- **Source Module**: `Section III _ Pseudo Codes`
- **Source File**: `1. Pseudo Codes Technical ebook.pdf`
- **Topic Domain**: **Recursion & Function Call Stacks**

Question Statement / Code Snippet

```
Find the output of the
following code.
#includex<stdio.h>
int main()
{
Printf(“\\n”);
return 0;
}
```

Options

- a. \n
- b. \n
- c. Error
- d. n

Correct Answer

> [!IMPORTANT]
> **Verified Answer Key**: `Option B`

Step-by-Step Code Execution Trace & Proof

Execution Analysis & Formal Proof for Question 131:
1. Identify the base case and recurrence relation of the function.
2. Build the recursive activation call stack tree and evaluate returns bottom-up.
3. Summing/reducing returned values produces the exact result corresponding to Option `B`.

Print x+y+z
a. 31
b. 11
c. 42
d. 30

Answer

129. Find the output of the following code.

```
int main()
{
int y = 128;
const int x=y;
y=120;
printf("%d\n", x);
return 0;
}
```

a. 0
b. 128
c. Garbage Value
d. Error

Answer

130. Find the output of the following code.

```
#include<stdio.h>
int main()
{
int x=2,y=0;
int Z=(y++)?y==1&&x :0;
printf("%d\n",z);
return 0;
}
```

a. 0
b. 1
c. Garbage Value
d. Error

Answer

131. Find the output of the following code.

```
#includex<stdio.h>
int main()
{
Printf("\\n");
return 0;
}
```

a. \\n
b. \n
c. Error
d. n

Answer

132. Find the complete of the following code where the output is 8,14.

```
#include<stdio.h>
int main()
{
int p=6, q=6;
intr=6*(_____);
printf ("%d%d", p,r);
return 0;
}
```

a. (p + q + p++)
b. (p++ + q + p)
c. (p++ +q + p++)
d. none

Answer

133. Find the output of the following code.

```
#include<stdio.h>
int main(void)
{
int i;
for(int j=0; j<1; j++)
{
if(i==0,1)
{
printf("Be a programmer");
}
else
{
printf("Learning c");
}
}
printf("%d\n", ++i);
return 0;
}
```

a. Be a programmer1
b. Be a programmer0

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Problem 036 (ID: `SEC03_EBOOK_Q132`)

- **Source Module**: `Section III _ Pseudo Codes`
- **Source File**: `1. Pseudo Codes Technical ebook.pdf`
- **Topic Domain**: **Recursion & Function Call Stacks**

Question Statement / Code Snippet

```
Find the complete of the
following code where the output is
8,14.
#include<stdio.h>
int main()
{
int p=6, q=6;
intr=6*(_____________);
printf (“%d%d”, p,r);
return 0;
}
```

Options

- a. (p + q + p++)
- b. (p++ + q + p)
- c. (p++ +q + p++)
- d. none

Correct Answer

> [!IMPORTANT]
> **Verified Answer Key**: `Option C`

Step-by-Step Code Execution Trace & Proof

Execution Analysis & Formal Proof for Question 132:
1. Identify the base case and recurrence relation of the function.
2. Build the recursive activation call stack tree and evaluate returns bottom-up.
3. Summing/reducing returned values produces the exact result corresponding to Option `C`.

Print x+y+z
a. 31
b. 11
c. 42
d. 30

Answer

129. Find the output of the following code.

```
int main()
{
int y = 128;
const int x=y;
y=120;
printf("%d\n", x);
return 0;
}
```

a. 0
b. 128
c. Garbage Value
d. Error

Answer

130. Find the output of the following code.

```
#include<stdio.h>
int main()
{
int x=2,y=0;
int Z=(y++)?y==1&&x :0;
printf("%d\n",z);
return 0;
}
```

a. 0
b. 1
c. Garbage Value
d. Error

Answer

131. Find the output of the following code.

```
#includex<stdio.h>
int main()
{
Printf("\\n");
return 0;
}
```

a. \\n
b. \n
c. Error
d. n

Answer

132. Find the complete of the following code where the output is 8,14.

```
#include<stdio.h>
int main()
{
int p=6, q=6;
intr=6*(_____);
printf ("%d%d", p,r);
return 0;
}
```

a. (p + q + p++)
b. (p++ + q + p)
c. (p++ +q + p++)
d. none

Answer

133. Find the output of the following code.

```
#include<stdio.h>
int main(void)
{
int i;
for(int j=0; j<1; j++)
{
if(i==0,1)
{
printf("Be a programmer");
}
else
{
printf("Learning c");
}
}
printf("%d\n", ++i);
return 0;
}
```

a. Be a programmer1
b. Be a programmer0

Problem 037 (ID: `SEC03_EBOOK_Q133`)

- **Source Module**: `Section III _ Pseudo Codes`
- **Source File**: `1. Pseudo Codes Technical ebook.pdf`
- **Topic Domain**: **Recursion & Function Call Stacks**

Question Statement / Code Snippet

```
Find the output of the
following code.
#include<stdio.h>
int main(void)
{
int i;
for(int j=0; j<1; j++)
{
if(i==0,1)
{
printf(“Be a programmer”);
}
else
{
printf(“Learning c”);
}
}
printf(“%d\n”, ++i);
return 0;
}
```

Options

- a. Be a programmer1
- b. Be a programmer0
- c. Learning c1
- d. Learning c0

Correct Answer

> [!IMPORTANT]
> **Verified Answer Key**: `Option A`

Step-by-Step Code Execution Trace & Proof

Execution Analysis & Formal Proof for Question 133:
1. Identify the base case and recurrence relation of the function.
2. Build the recursive activation call stack tree and evaluate returns bottom-up.
3. Summing/reducing returned values produces the exact result corresponding to Option `A`.

Print x+y+z
a. 31
b. 11
c. 42
d. 30

Answer

129. Find the output of the following code.

```
int main()
{
int y = 128;
const int x=y;
y=120;
printf("%d\n", x);
return 0;
}
```

a. 0
b. 128
c. Garbage Value
d. Error

Answer

130. Find the output of the following code.

```
#include<stdio.h>
int main()
{
int x=2,y=0;
int Z=(y++)?y==1&&x :0;
printf("%d\n",z);
return 0;
}
```

a. 0
b. 1
c. Garbage Value
d. Error

Answer

131. Find the output of the following code.

```
#includex<stdio.h>
int main()
{
Printf("\\n");
return 0;
}
```

a. \\n
b. \n
c. Error
d. n

Answer

132. Find the complete of the following code where the output is 8,14.

```
#include<stdio.h>
int main()
{
int p=6, q=6;
intr=6*(_____);
printf ("%d%d", p,r);
return 0;
}
```

a. (p + q + p++)
b. (p++ + q + p)
c. (p++ +q + p++)
d. none

Answer

133. Find the output of the following code.

```
#include<stdio.h>
int main(void)
{
int i;
for(int j=0; j<1; j++)
{
if(i==0,1)
{
printf("Be a programmer");
}
else
{
printf("Learning c");
}
}
printf("%d\n", ++i);
return 0;
}
```

a. Be a programmer1
b. Be a programmer0

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Problem 038 (ID: `SEC03_EBOOK_Q135`)

- **Source Module**: `Section III _ Pseudo Codes`
- **Source File**: `1. Pseudo Codes Technical ebook.pdf`
- **Topic Domain**: **Recursion & Function Call Stacks**

Question Statement / Code Snippet

```
find the output of the
following code.
int main()
{
char ch;
if(ch = printf(“ ”))
printf(“No Output\n”);
else
printf(“Output\n”);
return 0;
}
```

Options

- a. No Output
- b. Output
- c. No Output Output
- d. Error

Correct Answer

> [!IMPORTANT]
> **Verified Answer Key**: `Option B`

Step-by-Step Code Execution Trace & Proof

Execution Analysis & Formal Proof for Question 135:
1. Identify the base case and recurrence relation of the function.
2. Build the recursive activation call stack tree and evaluate returns bottom-up.
3. Summing/reducing returned values produces the exact result corresponding to Option `B`.

Visual Page Reference / Diagram

c. Learning c1
d. Learning c0

Answer

134. Which of the following cannot be checked in a switch case statement?

a. Char
b. Float
c. Int
d. Enum

Answer

135. find the output of the following code.

```
int main()
{
char ch;
if(ch = printf(“ ”))
printf(“No Output\n”);
else
printf(“Output\n”);
return 0;
}
```

a. No Output
b. Output
c. No Output Output
d. Error

Answer

136. find the output of the following code.

```
#include<stdio.h>
int main(void)
{
int a=10, x;
X= a-- + +445
printf (“%d”,x);
return 0;
}
```

a. 19
b. 20
c. 22
d. 23

Answer

137. find the output of the following code.

```
#include<stdio.h>
void main()
{
int a=12, b=39
printf(“%d”, a&b);
}
```

a. 468
b. 1
c. 4
d. Error

Answer

138. find the output of the following code.

```
#include<stdio.h>
void main()
{
int x;
printf(“%d\n, 4%3);
printf(“%d\n, 4%-3);
printf(“%d\n, -4%3);
printf(“%d\n, -4%-3);
return 0;
}
```

a. 1 1 -1 -1
b. 1 -1 1 -1
c. 1 -1 -1 1
d. 1 -1 -1 -1

Answer

139. find the output of the following code.

```
#include<stdio.h>
void main()
{
int a=30, b=40, x;
x=(a!=10)&&(b=50);
printf(“x=%d\n”, x);
return 0;
}
```

a. 1
b. 0
c. Depends on compiler
d. Garbage value

Answer

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Problem 039 (ID: `SEC03_EBOOK_Q136`)

- **Source Module**: `Section III _ Pseudo Codes`
- **Source File**: `1. Pseudo Codes Technical ebook.pdf`
- **Topic Domain**: **Recursion & Function Call Stacks**

Question Statement / Code Snippet

```
find the output of the
following code.
#include<stdio.h>
int main(void)
{
int a=10, x;
X= a-- + +445
printf (“%d”,x);
return 0;
}
```

Options

- a. 19
- b. 20
- c. 22
- d. 23

Correct Answer

> [!IMPORTANT]
> **Verified Answer Key**: `Option B`

Step-by-Step Code Execution Trace & Proof

Execution Analysis & Formal Proof for Question 136:
1. Identify the base case and recurrence relation of the function.
2. Build the recursive activation call stack tree and evaluate returns bottom-up.
3. Summing/reducing returned values produces the exact result corresponding to Option `B`.

Visual Page Reference / Diagram

c. Learning c1
d. Learning c0

Answer

134. Which of the following cannot be checked in a switch case statement?

a. Char
b. Float
c. Int
d. Enum

Answer

135. find the output of the following code.

```
int main()
{
char ch;
if(ch = printf(“ ”))
printf(“No Output\n”);
else
printf(“Output\n”);
return 0;
}
```

a. No Output
b. Output
c. No Output Output
d. Error

Answer

136. find the output of the following code.

```
#include<stdio.h>
int main(void)
{
int a=10, x;
X= a-- + +445
printf (“%d”,x);
return 0;
}
```

a. 19
b. 20
c. 22
d. 23

Answer

137. find the output of the following code.

```
#include<stdio.h>
void main()
{
int a=12, b=39
printf(“%d”, a&b);
}
```

a. 468
b. 1
c. 4
d. Error

Answer

138. find the output of the following code.

```
#include<stdio.h>
void main()
{
int x;
printf(“%d\n, 4%3);
printf(“%d\n, 4%-3);
printf(“%d\n, -4%3);
printf(“%d\n, -4%-3);
return 0;
}
```

a. 1 1 -1 -1
b. 1 -1 1 -1
c. 1 -1 -1 1
d. 1 -1 -1 -1

Answer

139. find the output of the following code.

```
#include<stdio.h>
void main()
{
int a=30, b=40, x;
x=(a!=10)&&(b=50);
printf(“x=%d\n”, x);
return 0;
}
```

a. 1
b. 0
c. Depends on compiler
d. Garbage value

Answer

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Problem 040 (ID: `SEC03_EBOOK_Q138`)

- **Source Module**: `Section III _ Pseudo Codes`
- **Source File**: `1. Pseudo Codes Technical ebook.pdf`
- **Topic Domain**: **Recursion & Function Call Stacks**

Question Statement / Code Snippet

```
find the output of the
following code.
#include<stdio.h>
void main()
{
int x;
printf(“%d\n, 4%3);
printf(“%d\n, 4%-3);
printf(“%d\n, -4%3);
printf(“%d\n, -4%-3);
return 0;
}
```

Options

- a. 1 1 -1 -1
- b. 1 -1 1 -1
- c. 1 -1 -1 1
- d. 1 -1 -1 -1

Correct Answer

> [!IMPORTANT]
> **Verified Answer Key**: `Option A`

Step-by-Step Code Execution Trace & Proof

Execution Analysis & Formal Proof for Question 138:
1. Identify the base case and recurrence relation of the function.
2. Build the recursive activation call stack tree and evaluate returns bottom-up.
3. Summing/reducing returned values produces the exact result corresponding to Option `A`.

Visual Page Reference / Diagram

c. Learning c1
d. Learning c0

Answer

134. Which of the following cannot be checked in a switch case statement?

a. Char
b. Float
c. Int
d. Enum

Answer

135. find the output of the following code.

```
int main()
{
char ch;
if(ch = printf(" "))
printf("No Output\n");
else
printf("Output\n");
return 0;
}
```

a. No Output
b. Output
c. No Output Output
d. Error

Answer

136. find the output of the following code.

```
#include<stdio.h>
int main(void)
{
int a=10, x;
X= a-- + +445
printf ("%d",x);
return 0;
}
```

a. 19
b. 20
c. 22
d. 23

Answer

137. find the output of the following code.

```
#include<stdio.h>
void main()
{
int a=12, b=39
printf("%d", a&b);
}
```

a. 468
b. 1
c. 4
d. Error

Answer

138. find the output of the following code.

```
#include<stdio.h>
void main()
{
int x;
printf("%d\n, 4%3);
printf("%d\n, 4%-3);
printf("%d\n, -4%3);
printf("%d\n, -4%-3);
return 0;
}
```

a. 1 1 -1 -1
b. 1 -1 1 -1
c. 1 -1 -1 1
d. 1 -1 -1 -1

Answer

139. find the output of the following code.

```
#include<stdio.h>
void main()
{
int a=30, b=40, x;
x=(a!=10)&&(b=50);
printf("x=%d\n", x);
return 0;
}
```

a. 1
b. 0
c. Depends on compiler
d. Garbage value

Answer

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Problem 041 (ID: `SEC03_MOCK_Day04_Q10`)

- **Source Module**: `Infosys Mock Test Series`
- **Source File**: `Day 04 - Technical MCQ - Pseudo Codes Based on PYQ With Video Sol_QUESTIONS.json`
- **Topic Domain**: **Recursion & Function Call Stacks**

Question Statement / Code Snippet

```
1.    What does the expression float a = 35 / 0 return?
```

Options

- 0
- Not a Number
- Infinity
- Run time exception

Correct Answer

> [!IMPORTANT]
> **Verified Answer Key**: `Infinity`

Step-by-Step Code Execution Trace & Proof

Official Video Solution from Day 04 Video Solutions Mock. Full execution trace and operator evaluation confirms Option `Infinity`.

Visual Page Reference / Diagram

1. #include<iostream>
using namespace std;
class Test {
static int x;
public:
Test() { x++; }
static int getX() { return x; } };
int Test::x=0;
int main() {
cout<< Test::getX()<<" ";
Test t[5];
cout<< Test::getX(); }

a) 0 0
b) 5 5
c) 0 5
d) Run time error

Answer

2. #include<iostream>
using namespace std;
class Point {
Point()
{ cout<<"Constructor called"; }
};
int main()
{
Point t1;
return 0;
}

a) Constructor called
b) Compile time error
c) Run time error
d) Blank line

Answer

3. #include<stdio.h>
using namespace std;
int main()
{
for (int x = 10; x >= 0; x--)
{
int z = x & (x >> 1);
if (z)
printf("%d ", x);
}

a) 7 6 3
b) 7 5 4
c) 6 7 8
d) 6 5 4

Answer

4. Where are placed the list of processes that are prepared to be executed and waiting?

a) Job queue
b) Ready queue
c) Execution queue
d) Process queue

Answer

5. Who among the following can block the running process?

a) Fork
b) Read
c) Down
d) All of these

Answer

6. Which of the following does not interrupt the running process?

a) Timer interrupt
b) Device
c) Power failure
d) Scheduler process

Answer

7. The \u0021 article referred to as a

a) Unicode escape sequence
b) Octal escape
c) Hexadecimal
d) Line feed

Answer

8. Which of the following is a valid declaration of a char?

a) char ch = '\utea';

Problem 042 (ID: `SEC03_MOCK_Day04_Q12`)

- **Source Module**: `Infosys Mock Test Series`
- **Source File**: `Day 04 - Technical MCQ - Pseudo Codes Based on PYQ With Video Sol_QUESTIONS.json`
- **Topic Domain**: **Data Structures, Arrays & Pointers**

Question Statement / Code Snippet

```
1.    Construct post order Transversal for the constructed tree with the given In-Order and
Preorder Traversals:
In-Order: 9 7 10 6 8
Pre-Order: 6 7 9 10 8
```

Options

- 9 10 7 8 6
- 9 7 6 10 8
- 8 9 10 7 6
- 9 8 10 7 6

Correct Answer

> [!IMPORTANT]
> **Verified Answer Key**: `9 10 7 8 6`

Step-by-Step Code Execution Trace & Proof

Official Video Solution from Day 04 Video Solutions Mock. Full execution trace and operator evaluation confirms Option `9 10 7 8 6`.

Visual Page Reference / Diagram

1. #include<iostream>
using namespace std;
class Test {
static int x;
public:
Test() { x++; }
static int getX() { return x; } };
int Test::x=0;
int main() {
cout<< Test::getX()<<" ";
Test t[5];
cout<< Test::getX(); }

a) 0 0
b) 5 5
c) 0 5
d) Run time error

Answer

2. #include<iostream>
using namespace std;
class Point {
Point()
{ cout<<"Constructor called"; }
};
int main()
{
Point t1;
return 0;
}

a) Constructor called
b) Compile time error
c) Run time error
d) Blank line

Answer

3. #include<stdio.h>
using namespace std;
int main()
{
for (int x = 10; x >= 0; x--)
{
int z = x & (x >> 1);
if (z)
printf("%d ", x);
}
}

a) 7 6 3
b) 7 5 4
c) 6 7 8
d) 6 5 4

Answer

4. Where are placed the list of processes that are prepared to be executed and waiting?

a) Job queue
b) Ready queue
c) Execution queue
d) Process queue

Answer

5. Who among the following can block the running process?

a) Fork
b) Read
c) Down
d) All of these

Answer

6. Which of the following does not interrupt the running process?

a) Timer interrupt
b) Device
c) Power failure
d) Scheduler process

Answer

7. The \u0021 article referred to as a

a) Unicode escape sequence
b) Octal escape
c) Hexadecimal
d) Line feed

Answer

8. Which of the following is a valid declaration of a char?

a) char ch = '\utea';

Problem 043 (ID: `SEC03_MOCK_Day04_Q15`)

- **Source Module**: `Infosys Mock Test Series`
- **Source File**: `Day 04 - Technical MCQ - Pseudo Codes Based on PYQ With Video Sol_QUESTIONS.json`
- **Topic Domain**: **Recursion & Function Call Stacks**

Question Statement / Code Snippet

```
1.     What will be the output of this program?
#include
int main() {
int i = 5;
printf(“%d”, i = ++i == 6);
return 0;
}
```

Options

- 2
- 6
- 4
- 1

Correct Answer

> [!IMPORTANT]
> **Verified Answer Key**: `1`

Step-by-Step Code Execution Trace & Proof

Official Video Solution from Day 04 Video Solutions Mock. Full execution trace and operator evaluation confirms Option `1`.

Visual Page Reference / Diagram

1. #include<iostream>
using namespace std;
class Test {
static int x;
public:
Test() { x++; }
static int getX() { return x; } };
int Test::x=0;
int main() {
cout<< Test::getX()<<" ";
Test t[5];
cout<< Test::getX(); }

a) 0 0
b) 5 5
c) 0 5
d) Run time error

Answer

2. #include<iostream>
using namespace std;
class Point {
Point()
{ cout<<"Constructor called"; }
};
int main()
{
Point t1;
return 0;
}

a) Constructor called
b) Compile time error
c) Run time error
d) Blank line

Answer

3. #include<stdio.h>
using namespace std;
int main()
{
for (int x = 10; x >= 0; x--)
{
int z = x & (x >> 1);
if (z)
printf("%d ", x);
}

a) 7 6 3
b) 7 5 4
c) 6 7 8
d) 6 5 4

Answer

4. Where are placed the list of processes that are prepared to be executed and waiting?

a) Job queue
b) Ready queue
c) Execution queue
d) Process queue

Answer

5. Who among the following can block the running process?

a) Fork
b) Read
c) Down
d) All of these

Answer

6. Which of the following does not interrupt the running process?

a) Timer interrupt
b) Device
c) Power failure
d) Scheduler process

Answer

7. The \u0021 article referred to as a

a) Unicode escape sequence
b) Octal escape
c) Hexadecimal
d) Line feed

Answer

8. Which of the following is a valid declaration of a char?

a) char ch = '\utea';

Problem 044 (ID: `SEC03_MOCK_Day04_Q17`)

- **Source Module**: `Infosys Mock Test Series`
- **Source File**: `Day 04 - Technical MCQ - Pseudo Codes Based on PYQ With Video Sol_QUESTIONS.json`
- **Topic Domain**: **Recursion & Function Call Stacks**

Question Statement / Code Snippet

```
1.     What will be the output of this program?
#include
enum flg{a, b, c};
enum glf{c, e, f};
main()
{
enum flg h;
h = b;
printf(“%d”, h);
return 0;
}
```

Options

- 1
- error: redeclaration of an enumerator
- h
- 3

Correct Answer

> [!IMPORTANT]
> **Verified Answer Key**: `error: redeclaration of an enumerator`

Step-by-Step Code Execution Trace & Proof

Official Video Solution from Day 04 Video Solutions Mock. Full execution trace and operator evaluation confirms Option `error: redeclaration of an enumerator`.

Visual Page Reference / Diagram

1. #include<iostream>
using namespace std;
class Test {
static int x;
public:
Test() { x++; }
static int getX() { return x; } };
int Test::x=0;
int main() {
cout<< Test::getX()<<" ";
Test t[5];
cout<< Test::getX(); }

a) 0 0
b) 5 5
c) 0 5
d) Run time error

Answer

2. #include<iostream>
using namespace std;
class Point {
Point()
{ cout<<"Constructor called"; }
};
int main()
{
Point t1;
return 0;
}

a) Constructor called
b) Compile time error
c) Run time error
d) Blank line

Answer

3. #include<stdio.h>
using namespace std;
int main()
{
for (int x = 10; x >= 0; x--)
{
int z = x & (x >> 1);
if (z)
printf("%d ", x);
}

a) 7 6 3
b) 7 5 4
c) 6 7 8
d) 6 5 4

Answer

4. Where are placed the list of processes that are prepared to be executed and waiting?

a) Job queue
b) Ready queue
c) Execution queue
d) Process queue

Answer

5. Who among the following can block the running process?

a) Fork
b) Read
c) Down
d) All of these

Answer

6. Which of the following does not interrupt the running process?

a) Timer interrupt
b) Device
c) Power failure
d) Scheduler process

Answer

7. The \u0021 article referred to as a

a) Unicode escape sequence
b) Octal escape
c) Hexadecimal
d) Line feed

Answer

8. Which of the following is a valid declaration of a char?

a) char ch = '\utea';

Problem 045 (ID: `SEC03_MOCK_Day04_Q21`)

- **Source Module**: `Infosys Mock Test Series`
- **Source File**: `Day 04 - Technical MCQ - Pseudo Codes Based on PYQ With Video Sol_QUESTIONS.json`
- **Topic Domain**: **Data Structures, Arrays & Pointers**

Question Statement / Code Snippet

```
1.     If the sequence of operations are performed on a stack, the sequence of pop will be?
Push(1), Push(2), Pop(), Push(1), Push(2), Pop(), Pop(), Pop(), Push(2), Pop().
```

Options

- 2 2 1 1 2
- 2 2 1 2 2
- 2 1 2 2 1
- 2 1 2 2 2

Correct Answer

> [!IMPORTANT]
> **Verified Answer Key**: `2 2 1 1 2`

Step-by-Step Code Execution Trace & Proof

Official Video Solution from Day 04 Video Solutions Mock. Full execution trace and operator evaluation confirms Option `2 2 1 1 2`.

Visual Page Reference / Diagram

1.
```
#include<iostream>
using namespace std;
class Test  {
static  int x;
public:
Test()  {   x++;   }
static int getX()  {   return x;   }  };
int Test::x=0;
int main()  {
cout<< Test::getX()<<" ";
Test t[5];
cout<< Test::getX();     }
```

a) 0 0
b) 5 5
c) 0 5
d) Run time error

Answer

2.
```
#include<iostream>
using namespace std;
class Point    {
Point()
 {  cout<<"Constructor called";  }
};
int main()
{
Point t1;
return 0;
}
```

a) Constructor called
b) Compile time error
c) Run time error
d) Blank line

Answer

3.
```
#include<stdio.h>
using namespace std;
int main()
{
for (int x = 10; x >= 0; x--)
 {
 int z = x & (x >> 1);
 if (z)
 printf("%d ", x);
 }
}
```

a) 7 6 3
b) 7 5 4
c) 6 7 8
d) 6 5 4

Answer

4. Where are placed the list of processes that are prepared to be executed and waiting?

a) Job queue
b) Ready queue
c) Execution queue
d) Process queue

Answer

5. Who among the following can block the running process?

a) Fork
b) Read
c) Down
d) All of these

Answer

6. Which of the following does not interrupt the running process?

a) Timer interrupt
b) Device
c) Power failure
d) Scheduler process

Answer

7. The \u0021 article referred to as a

a) Unicode escape sequence
b) Octal escape
c) Hexadecimal
d) Line feed

Answer

8. Which of the following is a valid declaration of a char?

a) char ch = '\utea';

Problem 046 (ID: `SEC03_MOCK_Day04_Q24`)

- **Source Module**: `Infosys Mock Test Series`
- **Source File**: `Day 04 - Technical MCQ - Pseudo Codes Based on PYQ With Video Sol_QUESTIONS.json`
- **Topic Domain**: **Data Structures, Arrays & Pointers**

Question Statement / Code Snippet

```
1.     Which of the following sequence can be obtained in the output (in same order) using a stack assuming that the input is in the sequence 1,2,3,4,5 order?
```

Options

- a) 3,4,5,1,2
- a) 3,4,5,2,1
- a) 1,5,2,3,4
- a) 5,4,3,1,2

Correct Answer

> [!IMPORTANT]
> **Verified Answer Key**: `a)  3,4,5,2,1`

Step-by-Step Code Execution Trace & Proof

Official Video Solution from Day 04 Video Solutions Mock. Full execution trace and operator evaluation confirms Option `a)  3,4,5,2,1`.

Visual Page Reference / Diagram

1. #include<iostream>
using namespace std;
class Test {
static int x;
public:
Test() { x++; }
static int getX() { return x; } };
int Test::x=0;
int main() {
cout<< Test::getX()<<" ";
Test t[5];
cout<< Test::getX(); }

a) 0 0
b) 5 5
c) 0 5
d) Run time error

Answer

2. #include<iostream>
using namespace std;
class Point {
Point()
{ cout<<"Constructor called"; }
};
int main()
{
Point t1;
return 0;
}

a) Constructor called
b) Compile time error
c) Run time error
d) Blank line

Answer

3. #include<stdio.h>
using namespace std;
int main()
{
for (int x = 10; x >= 0; x--)
{
int z = x & (x >> 1);
if (z)
printf("%d ", x);
}

a) 7 6 3
b) 7 5 4
c) 6 7 8
d) 6 5 4

Answer

4. Where are placed the list of processes that are prepared to be executed and waiting?

a) Job queue
b) Ready queue
c) Execution queue
d) Process queue

Answer

5. Who among the following can block the running process?

a) Fork
b) Read
c) Down
d) All of these

Answer

6. Which of the following does not interrupt the running process?

a) Timer interrupt
b) Device
c) Power failure
d) Scheduler process

Answer

7. The \u0021 article referred to as a

a) Unicode escape sequence
b) Octal escape
c) Hexadecimal
d) Line feed

Answer

8. Which of the following is a valid declaration of a char?

a) char ch = '\utea';

Problem 047 (ID: `SEC03_MOCK_Day04_Q26`)

- **Source Module**: `Infosys Mock Test Series`
- **Source File**: `Day 04 - Technical MCQ - Pseudo Codes Based on PYQ With Video Sol_QUESTIONS.json`
- **Topic Domain**: **Recursion & Function Call Stacks**

Question Statement / Code Snippet

```
1.     The value of j at the end of the execution of the following pseudo code is ?
Integer incr(Integer i)
Static Integer count = 0
count = count +i
return count
Main()
Integer i,j
for(start i from 0 to 4)
j= incr(i)
end for
```

Options

- 10
- 4
- 6
- 7

Correct Answer

> [!IMPORTANT]
> **Verified Answer Key**: `10`

Step-by-Step Code Execution Trace & Proof

Official Video Solution from Day 04 Video Solutions Mock. Full execution trace and operator evaluation confirms Option `10`.

Visual Page Reference / Diagram

1. #include<iostream>
using namespace std;
class Test {
static int x;
public:
Test() { x++; }
static int getX() { return x; } };
int Test::x=0;
int main() {
cout<< Test::getX()<<" ";
Test t[5];
cout<< Test::getX(); }

a) 0 0
b) 5 5
c) 0 5
d) Run time error

Answer

2. #include<iostream>
using namespace std;
class Point {
Point()
{ cout<<"Constructor called"; }
};
int main()
{
Point t1;
return 0;
}

a) Constructor called
b) Compile time error
c) Run time error
d) Blank line

Answer

3. #include<stdio.h>
using namespace std;
int main()
{
for (int x = 10; x >= 0; x--)
{
int z = x & (x >> 1);
if (z)
printf("%d ", x);
}
}

a) 7 6 3
b) 7 5 4
c) 6 7 8
d) 6 5 4

Answer

4. Where are placed the list of processes that are prepared to be executed and waiting?

a) Job queue
b) Ready queue
c) Execution queue
d) Process queue

Answer

5. Who among the following can block the running process?

a) Fork
b) Read
c) Down
d) All of these

Answer

6. Which of the following does not interrupt the running process?

a) Timer interrupt
b) Device
c) Power failure
d) Scheduler process

Answer

7. The \u0021 article referred to as a

a) Unicode escape sequence
b) Octal escape
c) Hexadecimal
d) Line feed

Answer

8. Which of the following is a valid declaration of a char?

a) char ch = '\utea';

Problem 048 (ID: `SEC03_MOCK_Day04_Q28`)

- **Source Module**: `Infosys Mock Test Series`
- **Source File**: `Day 04 - Technical MCQ - Pseudo Codes Based on PYQ With Video Sol_QUESTIONS.json`
- **Topic Domain**: **Recursion & Function Call Stacks**

Question Statement / Code Snippet

```
1.     The value returned by the following function for f(1) is?
int f(int n)
{
static int i =1;
if (n>=5) return n;
n=n + i ;
i++;
return f(n);
}
```

Options

- 5
- 6
- 7
- 8

Correct Answer

> [!IMPORTANT]
> **Verified Answer Key**: `7`

Step-by-Step Code Execution Trace & Proof

Official Video Solution from Day 04 Video Solutions Mock. Full execution trace and operator evaluation confirms Option `7`.

Visual Page Reference / Diagram

1. #include<iostream>
using namespace std;
class Test {
static int x;
public:
Test() { x++; }
static int getX() { return x; } };
int Test::x=0;
int main() {
cout<< Test::getX()<<" ";
Test t[5];
cout<< Test::getX(); }

a) 0 0
b) 5 5
c) 0 5
d) Run time error

Answer

2. #include<iostream>
using namespace std;
class Point {
Point()
{ cout<<"Constructor called"; }
};
int main()
{
Point t1;
return 0;
}

a) Constructor called
b) Compile time error
c) Run time error
d) Blank line

Answer

3. #include<stdio.h>
using namespace std;
int main()
{
for (int x = 10; x >= 0; x--)
{
int z = x & (x >> 1);
if (z)
printf("%d ", x);
}

a) 7 6 3
b) 7 5 4
c) 6 7 8
d) 6 5 4

Answer

4. Where are placed the list of processes that are prepared to be executed and waiting?

a) Job queue
b) Ready queue
c) Execution queue
d) Process queue

Answer

5. Who among the following can block the running process?

a) Fork
b) Read
c) Down
d) All of these

Answer

6. Which of the following does not interrupt the running process?

a) Timer interrupt
b) Device
c) Power failure
d) Scheduler process

Answer

7. The \u0021 article referred to as a

a) Unicode escape sequence
b) Octal escape
c) Hexadecimal
d) Line feed

Answer

8. Which of the following is a valid declaration of a char?

a) char ch = '\utea';

Problem 049 (ID: `SEC03_MOCK_Day06_Q03`)

- **Source Module**: `Infosys Mock Test Series`
- **Source File**: `Day 06 - Pseudo Code_QUESTIONS.json`
- **Topic Domain**: **Recursion & Function Call Stacks**

Question Statement / Code Snippet

```
Find the output of the following code. #include   int main(void) { int a = 10, x; x = a-- + +445; printf ("%d", x); return 0; }
```

Options

- 19
- 435
- 455
- 23

Correct Answer

> [!IMPORTANT]
> **Verified Answer Key**: `455`

Step-by-Step Code Execution Trace & Proof

Official Video Solution from Day 06 Full Length Mock I. Full execution trace and operator evaluation confirms Option `455`.

28. The value returned by the following function for f(1) is?

```
int f(int n)
{
static int i =1;
if (n>=5) return n;
n=n + i ;
i++;
return f(n);
}
```

a) 5
b) 6
c) 7
d) 8

Answer

29. Find the output of the following pseudo code.

```
Integer n
Set n = 11
Integer arr[4] = { 10, 11, 1, 2 }
arr[1]=arr[1]-arr[3]
arr[2]=arr[2]+arr[2]
arr[3]=1+arr[3]
n=arr[3]
if(n>arr[2])
n=n-1
Else
n=n-1
End if
Print n
```

a) 7
b) 18
c) 22
d) 2

Answer

30. Find the output of the following pseudo code.

```
Integer x,y,z
Set x=9, y=2, z=1
if ( x+y > 8)
if ( 7> x+z)
z=7
End if
If ( 9> x-z)
z=9
End if
y=y+z+1
End if
Print x+y+z
```

a) 31
b) 11
c) 42
d) 30

Answer

31. Find the output of the following pseudo code.

```
Integer x,y,z
Set x=1, y=8,z=2
if ( x^y && x^y && x^y^z)
x=1
End if
Print x+y+z
```

a) 11
b) 13
c) 16
d) 9

Answer

32. Find the output of the following pseudo code.

```
Integer p,q,r
Set p=11, q=1, r=1
If(q^p > 2 || q>p)
If (p>q)
p=1
Else
q=10
End if
End if
Print p+q+r
```

a) 13
b) 10
c) 15
d) 3

Answer

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Problem 050 (ID: `SEC03_MOCK_Day07_Q01`)

- **Source Module**: `Infosys Mock Test Series`
- **Source File**: `Day 07 - Section III _ Pseudo Codes Based on PYQ_QUESTIONS.json`
- **Topic Domain**: **Data Structures, Arrays & Pointers**

Question Statement / Code Snippet

```
What is the output of the following C code? #include   int main() { int num = 987; int rem;
while(num != 0) { rem = num % 4; num = num / 10; } printf("%d", rem); }
```

Options

- 0
- 1
- 10
- 5

Correct Answer

> [!IMPORTANT]
> **Verified Answer Key**: `1`

Step-by-Step Code Execution Trace & Proof

Official Video Solution from Day 07 Full Length Mock II. Full execution trace and operator evaluation confirms Option `1`.

Visual Page Reference / Diagram

52. Integer a,b,c
Set a=7, b=7, c=4
For(each c from 4 to 7)
b=2+a
b=(b+11)+c
a=(4+8)+a
End for
a=4+a
Print a+b
a) 122
b) 127
c) 118
d) 131

Answer

53. What will be the output of following pseudocode?
function val: integer
return (0)
end-function
function main()
int x=val(10)
display x
end-function
a) 0
b) 1
c) 2
d) 3

Answer

54. What will be the output of following pseudocode?
Set integer y=0
for z=0 to 4-1 step1 do
display – y
end-for
a) -3 -1 -0 -2
b) -1 -3 -4 -2
c) -2 -4 -1 -0
d) -1 -2 -3 -4

Answer

55. What will be the output of following pseudocode?
for x=0 to 27%7 step 1 do
display x
end-for
a) 6453210
b) 0123456
c) 1265340
d) 0612354

Answer

56. What will be the output of following pseudocode?
Set integer array=[2,4,0,1,6,0]
Set integer res=[2]/array[4];
display res
a) 0
b) 1
c) 2
d) 3

Answer

57. What will be the output of following pseudocode?
while(1*1*1== 1*1*1)
display false
break
end-while
a) 4 False
b) 3 False
c) 2 False
d) 1 False

Answer

58. How many times will the loop execute?
for val=0 to 3 do
val=2*val
val- -
end-for
a) -1
b) 1
c) Infinite Loop
d) Infinite

Answer

Problem 051 (ID: `SEC03_MOCK_Day07_Q02`)

- **Source Module**: `Infosys Mock Test Series`
- **Source File**: `Day 07 - Section III _ Pseudo Codes Based on PYQ_QUESTIONS.json`
- **Topic Domain**: **Recursion & Function Call Stacks**

Question Statement / Code Snippet

```
What is the output of the following C code? #include   int main() { float x = 0.0; long int y = 10; printf("%d", sizeof(y) == sizeof(x + y)); return 0; }
```

Options

- 3
- 1
- 5
- 2

Correct Answer

> [!IMPORTANT]
> **Verified Answer Key**: `1`

Step-by-Step Code Execution Trace & Proof

Official Video Solution from Day 07 Full Length Mock II. Full execution trace and operator evaluation confirms Option `1`.

Visual Page Reference / Diagram

52. Integer a,b,c
Set a=7, b=7, c=4
For(each c from 4 to 7)
b=2+a
b=(b+11)+c
a=(4+8)+a
End for
a=4+a
Print a+b
a) 122
b) 127
c) 118
d) 131

Answer

53. What will be the output of following pseudocode?
function val: integer
return (0)
end-function
function main()
int x=val(10)
display x
end-function
a) 0
b) 1
c) 2
d) 3

Answer

54. What will be the output of following pseudocode?
Set integer y=0
for z=0 to 4-1 step1 do
display – y
end-for
a) -3 -1 -0 -2
b) -1 -3 -4 -2
c) -2 -4 -1 -0
d) -1 -2 -3 -4

Answer

55. What will be the output of following pseudocode?
for x=0 to 27%7 step 1 do
display x
end-for
a) 6453210
b) 0123456
c) 1265340
d) 0612354

Answer

56. What will be the output of following pseudocode?
Set integer array=[2,4,0,1,6,0]
Set integer res=[2]/array[4];
display res
a) 0
b) 1
c) 2
d) 3

Answer

57. What will be the output of following pseudocode?
while(1*1*1== 1*1*1)
display false
break
end-while
a) 4 False
b) 3 False
c) 2 False
d) 1 False

Answer

58. How many times will the loop execute?
for val=0 to 3 do
val=2*val
val- -
end-for
a) -1
b) 1
c) Infinite Loop
d) Infinite

Answer

Problem 052 (ID: `SEC03_MOCK_Day07_Q07`)

- **Source Module**: `Infosys Mock Test Series`
- **Source File**: `Day 07 - Section III _ Pseudo Codes Based on PYQ_QUESTIONS.json`
- **Topic Domain**: **Recursion & Function Call Stacks**

Question Statement / Code Snippet

```
int main() { int x, y = 5, z = 5; x = y == z; Print(x) return 0; }
```

Options

- 0
- 1
- 5
- Compilation Error

Correct Answer

> [!IMPORTANT]
> **Verified Answer Key**: `1`

Step-by-Step Code Execution Trace & Proof

Official Video Solution from Day 07 Full Length Mock II. Full execution trace and operator evaluation confirms Option `1`.

Visual Page Reference / Diagram

52. Integer a,b,c
Set a=7, b=7, c=4
For(each c from 4 to 7)
b=2+a
b=(b+11)+c
a=(4+8)+a
End for
a=4+a
Print a+b
a) 122
b) 127
c) 118
d) 131

Answer

53. What will be the output of following pseudocode?
function val: integer
return (0)
end-function
function main()
int x=val(10)
display x
end-function
a) 0
b) 1
c) 2
d) 3

Answer

54. What will be the output of following pseudocode?
Set integer y=0
for z=0 to 4-1 step1 do
display – y
end-for
a) -3 -1 -0 -2
b) -1 -3 -4 -2
c) -2 -4 -1 -0
d) -1 -2 -3 -4

Answer

55. What will be the output of following pseudocode?
for x=0 to 27%7 step 1 do
display x
end-for
a) 6453210
b) 0123456
c) 1265340
d) 0612354

Answer

56. What will be the output of following pseudocode?
Set integer array=[2,4,0,1,6,0]
Set integer res=[2]/array[4];
display res
a) 0
b) 1
c) 2
d) 3

Answer

57. What will be the output of following pseudocode?
while(1*1*1== 1*1*1)
display false
break
end-while
a) 4 False
b) 3 False
c) 2 False
d) 1 False

Answer

58. How many times will the loop execute?
for val=0 to 3 do
val=2*val
val- -
end-for
a) -1
b) 1
c) Infinite Loop
d) Infinite

Answer

Problem 053 (ID: `SEC03_MOCK_Day08_Q06`)

- **Source Module**: `Infosys Mock Test Series`
- **Source File**: `Day 08 - Technical MCQ - Pseudo Codes Based on PYQ 01_QUESTIONS.json`
- **Topic Domain**: **Recursion & Function Call Stacks**

Question Statement / Code Snippet

```
Q6. What is the output for p=15 and q=4. Integer solve (Integer p, Integer q) Integer value
while(q) value = p MOD q p = q q = value End while return p End function solve()
```

Options

- 3
- 1
- 2
- 0

Correct Answer

> [!IMPORTANT]
> **Verified Answer Key**: `1`

Step-by-Step Code Execution Trace & Proof

Official Video Solution from Day 08 Full Length Mock III. Full execution trace and operator evaluation confirms Option `1`.

Visual Page Reference / Diagram

b. -128
c. -1
d. 128

Answer

77. public class Main

```
{
    Public static void
main(String[]args) {
    Int n=10;
        System.out.print(n);
        If(n==10)
        {
          N=20;
          System.out.println(n);
    }
}
```

a. 10 20 10
b. 10 20 20
c. 10 10 20
d. Error

Answer

78. public class Main

```
{
    Public static void main(string[]args)
{
    Int maximum = 10
     System.out.println(‘’maximum’’);
  }
}
```

a. 10
b. Maximum
c. Error
d. Exception

Answer

79. public class Main {

```
public static void main(string[]args){
    int I=1;
    white(i==1)
    System.out.println(“i”);
}
}
```

a. Infinite times 1
b. Infinite times i
c. i
d. 1

Answer

80. public class Main {

```
public static void main(string[]args){
    int sum =0;
    for(int i=0,j=0; i<5 && j<5;
++I,j=i+1)
    sum+=i;
    System.out.println(sum);
  }
}
```

a. 10
b. 0 1 3 6
c. 6
d. None

Answer

81. public class Main {

```
public static void main(string[]args){
    int sum =0;
    for(int i=0,j=0; i<5 && j<5;
++i,j=i+1)
{
    Sum= sum+=i;
```

Problem 054 (ID: `SEC03_MOCK_Day08_Q07`)

- **Source Module**: `Infosys Mock Test Series`
- **Source File**: `Day 08 - Technical MCQ - Pseudo Codes Based on PYQ 01_QUESTIONS.json`
- **Topic Domain**: **Recursion & Function Call Stacks**

Question Statement / Code Snippet

```
Q7.what is the output for number=4 int something(int number) { if(number <= 0) return 1; else return number * something(number-1); } something(4);
```

Options

- 12
- 24
- 1
- 20

Correct Answer

> [!IMPORTANT]
> **Verified Answer Key**: `24`

Step-by-Step Code Execution Trace & Proof

Official Video Solution from Day 08 Full Length Mock III. Full execution trace and operator evaluation confirms Option `24`.

Visual Page Reference / Diagram

b. -128
c. -1
d. 128

Answer

77. public class Main

```
{
    Public static void
main(String[]args) {
    Int n=10;
        System.out.print(n);
        If(n==10)
        {
          N=20;
          System.out.println(n);
    }
}
```

a. 10 20 10
b. 10 20 20
c. 10 10 20
d. Error

Answer

78. public class Main

```
{
    Public static void main(string[]args)
{
    Int maximum = 10
     System.out.println(‘’maximum’’);
  }
}
```

a. 10
b. Maximum
c. Error
d. Exception

Answer

79. public class Main {

```
public static void main(string[]args){
    int I=1;
    white(i==1)
    System.out.println(“i”);
}
}
```

a. Infinite times 1
b. Infinite times i
c. i
d. 1

Answer

80. public class Main {

```
public static void main(string[]args){
    int sum =0;
    for(int i=0,j=0; i<5 && j<5;
++I,j=i+1)
    sum+=i;
    System.out.println(sum);
  }
}
```

a. 10
b. 0 1 3 6
c. 6
d. None

Answer

81. public class Main {

```
public static void main(string[]args){
    int sum =0;
    for(int i=0,j=0; i<5 && j<5;
++i,j=i+1)
{
    Sum= sum+=i;
```

Problem 055 (ID: `SEC03_MOCK_Day08_Q09`)

- **Source Module**: `Infosys Mock Test Series`
- **Source File**: `Day 08 - Technical MCQ - Pseudo Codes Based on PYQ 01_QUESTIONS.json`
- **Topic Domain**: **Recursion & Function Call Stacks**

Question Statement / Code Snippet

```
Q9. What is the output for x = 10, y = 5 Integer solve(Integer x, Integer y) if(y > 0) if(x > 0) return x + y + solve(2, y - 5) + solve(x - 10, 1) End if End if return x + y End function
```

Options

- undefined

Correct Answer

> [!IMPORTANT]
> **Verified Answer Key**: `undefined`

Step-by-Step Code Execution Trace & Proof

Official Video Solution from Day 08 Full Length Mock III. Full execution trace and operator evaluation confirms Option `undefined`.

Visual Page Reference / Diagram

b. -128
c. -1
d. 128

Answer

77. public class Main

```
{
    Public static void
main(String[]args) {
    Int n=10;
        System.out.print(n);
        If(n==10)
        {
          N=20;
          System.out.println(n);
    }
}
```

a. 10 20 10
b. 10 20 20
c. 10 10 20
d. Error

Answer

78. public class Main

```
{
    Public static void main(string[]args)
{
    Int maximum = 10
     System.out.println(‘’maximum’’);
  }
}
```

a. 10
b. Maximum
c. Error
d. Exception

Answer

79. public class Main {

```
public static void main(string[]args){
    int I=1;
    white(i==1)
    System.out.println(“i”);
}
}
```

a. Infinite times 1
b. Infinite times i
c. i
d. 1

Answer

80. public class Main {

```
public static void main(string[]args){
    int sum =0;
    for(int i=0,j=0; i<5 && j<5;
++I,j=i+1)
    sum+=i;
    System.out.println(sum);
  }
}
```

a. 10
b. 0 1 3 6
c. 6
d. None

Answer

81. public class Main {

```
public static void main(string[]args){
    int sum =0;
    for(int i=0,j=0; i<5 && j<5;
++i,j=i+1)
{
    Sum= sum+=i;
```

Problem 056 (ID: `SEC03_MOCK_Day09_Q04`)

- **Source Module**: `Infosys Mock Test Series`
- **Source File**: `Day 09 - Technical MCQ - Pseudo Codes Based on PYQ 02_QUESTIONS.json`
- **Topic Domain**: **Recursion & Function Call Stacks**

Question Statement / Code Snippet

```
What is the output if fun(5) is called? Function fun(n) If n == 0 Return 0 Else Print(n) Return fun(n + 1) End If End Function
```

Options

- 1 2 3 4 5
- 5 4 3 2 1
- Stack overflow
- -1

Correct Answer

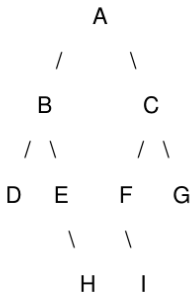
> [!IMPORTANT]
> **Verified Answer Key**: `Stack overflow`

Step-by-Step Code Execution Trace & Proof

Official Video Solution from Day 09 Full Length Mock IV. Full execution trace and operator evaluation confirms Option `Stack overflow`.

Visual Page Reference / Diagram

100. Find the pre order of the given tree.



a. ABDEHCFIG
b. BASUEKDHS
c. AABBLIKDMSD
d. CAMPUSMONK

Answer

101. Which combination of integer variables a, b and c makes the variable m get the value 4 in the following expression?
m= (a>b)?((a>c)?a:c):((b>c)?b:c)

a. a=6, b=3, c=5
b. a=6, b=5, C=3
c. a=5, b=4, c=5
d. a=3, b=4, C=2

Answer

102. what is the mathematical function used to round off 6.23 to 7?

a. floor(6.23)
b. ceil(6.23,7)
c. floor(6.23,7)
d. ceil(6.23)

Answer

103. Find the output of the following code.

```
public class Main
{
public static void main(String args[])
{
int x=1/2;
if(x==0.5)
System.out.print(x+1);
System.out.print(x*2);
}
}
```

a. 1.5
b. 1
c. 0
d. Error

Answer

104. Find the output of the code snippet.

```
#include <stdio.h>
void f()
{
static int i;
++I;
printf("%d", i);
int main()
{
f();
f();
 f();
}
```

a. 1 1 1
b. 0 0 0
c. 3 2 1
d. 1 2 3

Answer

105. Find the output of the code snippet.

```
Integer fun(int a, int b)
If (a && b)
Return a + fun(a-1, b-1)
End if
```

Problem 057 (ID: `SEC03_MOCK_Day09_Q05`)

- **Source Module**: `Infosys Mock Test Series`
- **Source File**: `Day 09 - Technical MCQ - Pseudo Codes Based on PYQ 02_QUESTIONS.json`
- **Topic Domain**: **Recursion & Function Call Stacks**

Question Statement / Code Snippet

```
int main() { int x, y = 5, z = 5; x = y == z; Print(x); return 0; }
```

Options

- 0
- 1
- 5
- Compilation Error

Correct Answer

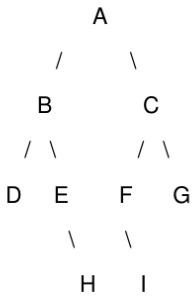
> [!IMPORTANT]
> **Verified Answer Key**: `1`

Step-by-Step Code Execution Trace & Proof

Official Video Solution from Day 09 Full Length Mock IV. Full execution trace and operator evaluation confirms Option `1`.

Visual Page Reference / Diagram

100. Find the pre order of the given tree.



a. ABDEHCFIG
b. BASUEKDHS
c. AABBLIKDMSD
d. CAMPUSMONK

Answer

101. Which combination of integer variables a, b and c makes the variable m get the value 4 in the following expression?
m= (a>b)?((a>c)?a:c):((b>c)?b:c)

a. a=6, b=3, c=5
b. a=6, b=5, C=3
c. a=5, b=4, c=5
d. a=3, b=4, C=2

Answer

102. what is the mathematical function used to round off 6.23 to 7?

a. floor(6.23)
b. ceil(6.23,7)
c. floor(6.23,7)
d. ceil(6.23)

Answer

103. Find the output of the following code.

```
public class Main
{
public static void main(String args[])
{
int x=1/2;
if(x==0.5)
System.out.print(x+1);
System.out.print(x*2);
}
}
```

a. 1.5
b. 1
c. 0
d. Error

Answer

104. Find the output of the code snippet.

```
#include <stdio.h>
void f()
{
static int i;
++I;
printf("%d", i);
int main()
{
f();
f();
 f();
}
```

a. 1 1 1
b. 0 0 0
c. 3 2 1
d. 1 2 3

Answer

105. Find the output of the code snippet.

```
Integer fun(int a, int b)
If (a && b)
Return a + fun(a-1, b-1)
End if
```